

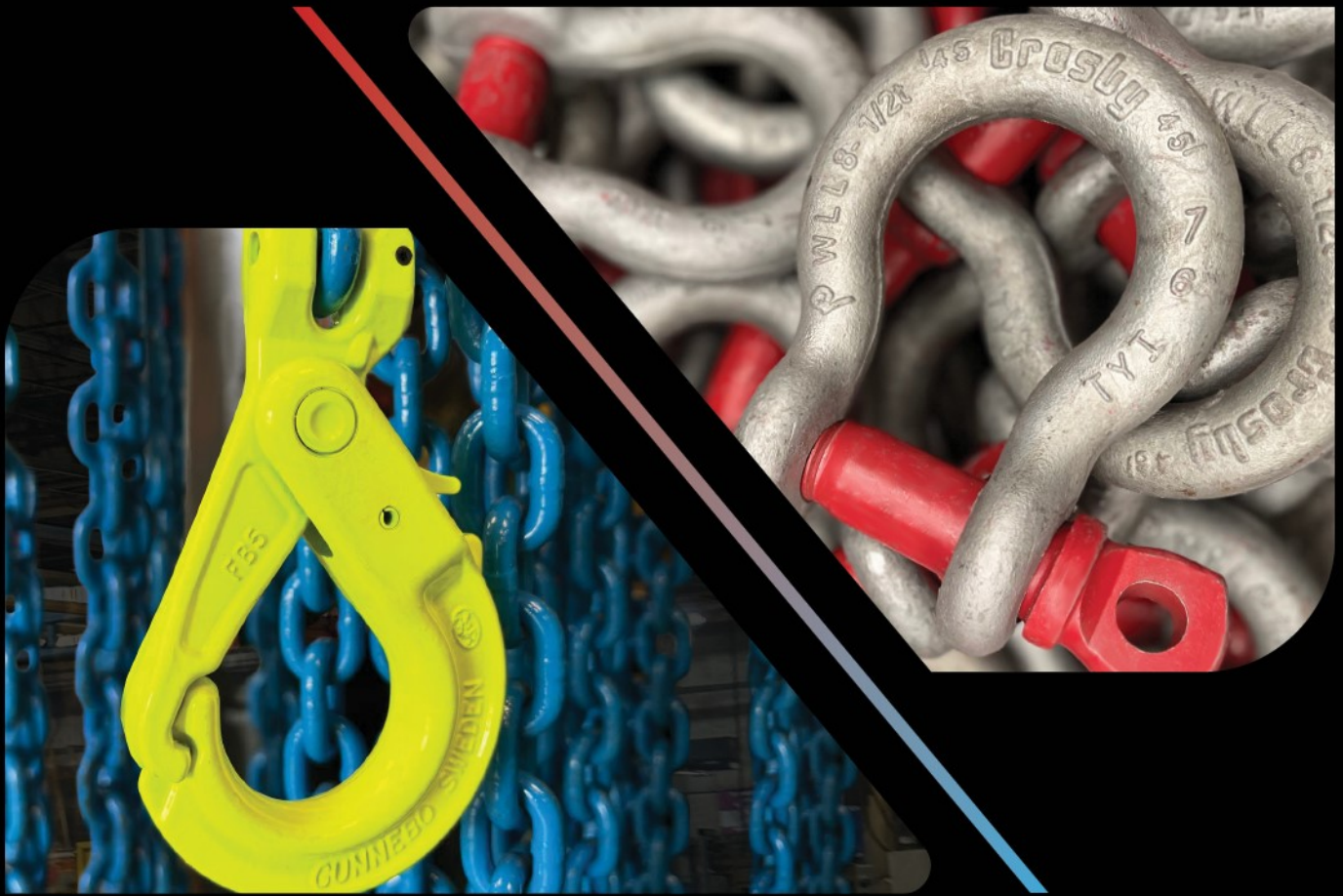


TURNBUCKLES



CoreLifting.com





Crosby®

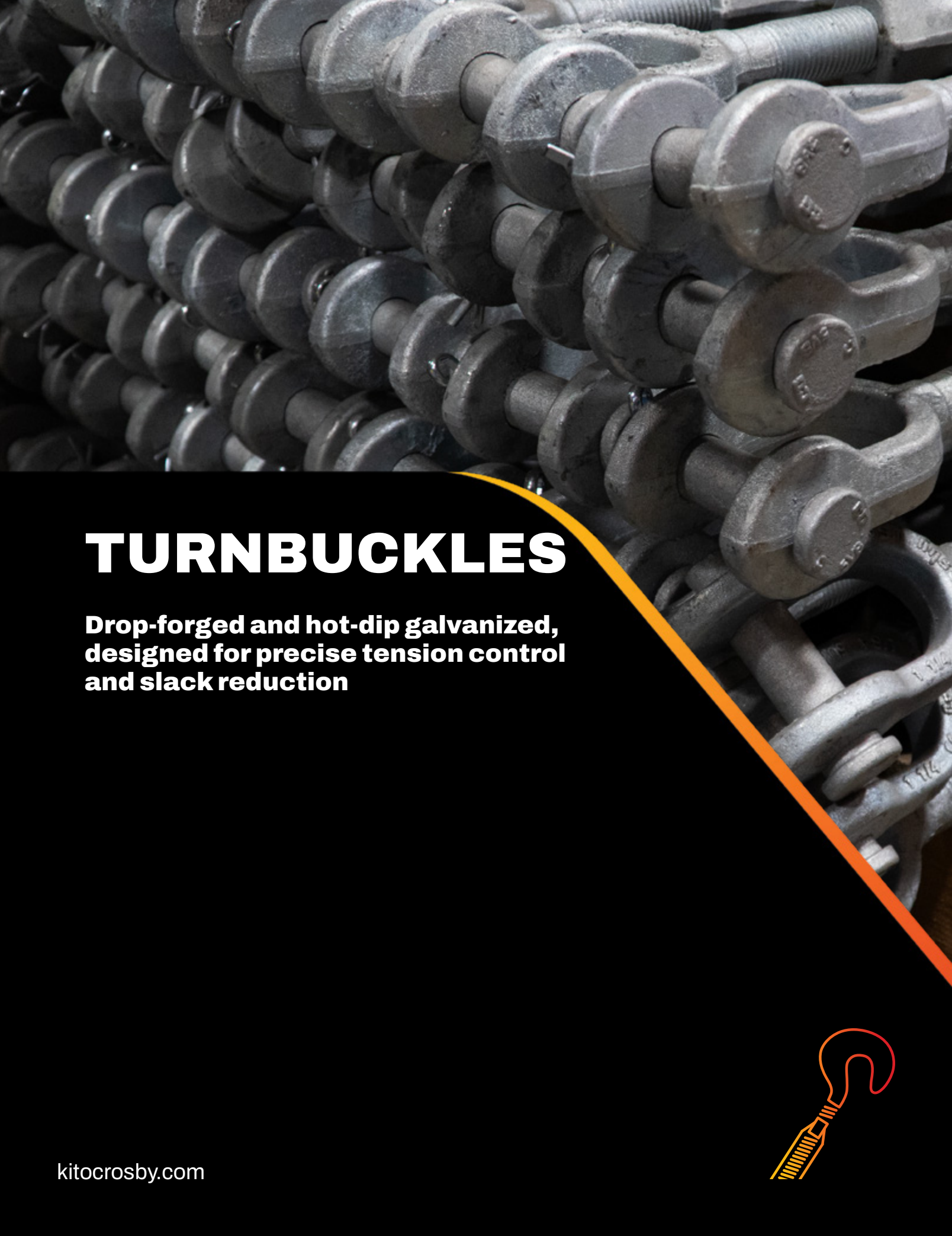


GUNNEBO®
Industries

PRODUCT CATALOG

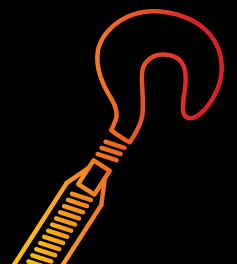
with application & warning information

Imperial



TURNBUCKLES

**Drop-forged and hot-dip galvanized,
designed for precise tension control
and slack reduction**



HG-223

HOOK & HOOK

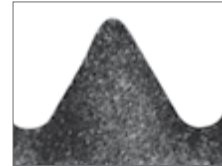
Meets the performance requirements of Federal Specifications FF-T-791b, Type 1, Form 1, Class 5, and ASTM F-1145, except for those provisions required of the contractor.



HG-226

EYE & EYE

Meets the performance requirements of Federal Specifications FF-T-791b, Type 1, Form 1, Class 4, and ASTM F-1145, except for those provisions required of the contractor.



Modified Thread:
Note stress relieving radii in this unretouched photo enlargement of the supabuckle.

HG-227

JAW & EYE

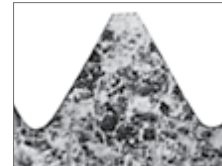
Meets the performance requirements of Federal Specifications FF-T-791b, Type 1, Form 1, Class 8, and ASTM F-1145, except for those provisions required of the contractor.



HG-228

JAW & JAW

Meets the performance requirements of Federal Specifications FF-T-791b, Type 1, Form 1, Class 7, and ASTM F-1145, except for those provisions required of the contractor.



Standard Thread:
Note stress building sharp "V" in this untouched photo enlargement.

Turnbuckle Information

- Turnbuckle assembly combinations include: eye & eye, hook & hook, jaw & jaw, and jaw & eye.
- End fittings are Quenched & Tempered or normalized, bodies heat treated by normalizing.
- Crosby Quenched & Tempered end fittings and normalized bodies have enhanced impact properties for greater toughness at all temperatures.
- Hot-dip galvanized.
- Hooks are forged with a greater cross sectional area that results in a stronger hook with better fatigue properties.
- Modified UNJ thread on end fittings for improved fatigue properties. Body has UNC threads.
- Turnbuckle eyes are forged elongated, by design, to maximize easy attachment in system and minimize stress in the eye. For turnbuckle sizes 1/4" through 2-1/2", a shackle one size smaller can be reeved through eye.
- Forged jaw ends are fitted with bolts and nuts on size 1/4" - 5/8", and pins and cotter on sizes 3/4" through 2-3/4".
- Can be used in general service conditions down to temperatures of -40° F (-40° C) and up to 400° F (204° C). See page 3 for more details.
- TURNBUCKLES RECOMMENDED FOR STRAIGHT OR IN-LINE PULL ONLY.
- Lock nuts available for all sizes.
- Typical hardness levels, tensile strengths and ductility properties are available for all sizes.
- Meets or exceeds all the requirements of ASME B30.26 including identification, ductility, design factor, proof load and temperature requirements. Importantly, these turnbuckles meet other critical performance requirements, including fatigue life, impact properties and material traceability, not addressed by ASME B30.26.

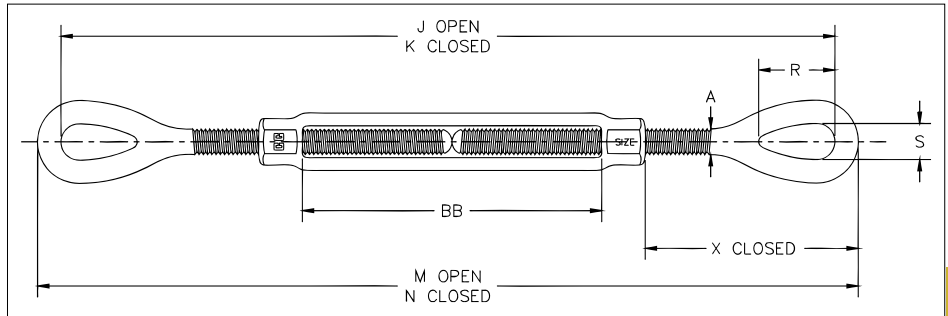


HG-226



APPLICATION AND WARNING INFORMATION
SECTION 17

- End fittings are Quenched & Tempered or normalized, bodies heat-treated by normalizing.
- Hot-dip galvanized steel.
- Turnbuckle eyes are forged elongated, by design, to maximize easy attachment in system and minimize stress in the eye. For turnbuckle sizes 1/4" through 2-1/2", a shackle one size smaller can be reeved through eye.
- Modified UNJ thread on end fittings for improved fatigue properties. Body has UNC threads.
- TURNBUCKLES RECOMMENDED FOR STRAIGHT OR IN-LINE PULL ONLY.
- Lock nuts available for all sizes.
- Fatigue rated to 20,000 cycles at 1-1/2 times the Working Load Limit.
- Meets or exceeds all requirements of ASME B30.26 including identification, ductility, design factor, proof load, and temperature requirements. Importantly, these turnbuckles meet other critical performance requirements including fatigue life, impact properties, and material traceability, not addressed by ASME B30.26.
- Meets the performance requirements of Federal Specifications FF-T-791b, Type 1 Form 1 - CLASS 4, and ASTM F-1145, except for those provisions required of the contractor. For additional information, see the warnings and applications section.



HG-226 Eye & Eye

Thread Dia. & Take Up (in)	Stock No.	Working Load Limit (lb)	Weight Each (lb)	Dimensions (in)								
				A	J Open	K Closed	M Open	N Closed	R	S	X Closed	BB
* 5/16 x 4-1/2	1031270	800	.48	.31	13.92	9.42	14.48	9.98	.95	.44	2.20	4.58
* 3/8 x 6	1031298	1200	.75	.38	17.56	11.56	18.24	12.24	1.13	.53	2.48	6.10
1/2 x 6	1031314	2200	1.72	.50	19.94	13.94	20.82	14.82	1.41	.71	3.56	6.03
1/2 x 12	1031350	2200	2.63	.50	32.23	20.23	33.11	21.11	1.41	.71	3.54	12.36
5/8 x 6	1031378	3500	2.75	.63	21.72	15.72	22.72	16.72	1.80	.88	4.35	6.03
5/8 x 12	1031412	3500	4.12	.63	34.06	22.06	35.06	23.06	1.80	.88	4.34	12.39
3/4 x 6	1031430	5200	4.22	.75	23.24	17.24	24.50	18.50	2.09	1.00	5.12	6.13
3/4 x 12	1031476	5200	6.12	.75	35.64	23.64	36.90	24.90	2.09	1.00	5.09	12.59
3/4 x 18	1031494	5200	7.83	.75	47.64	29.64	48.90	30.90	2.09	1.00	5.12	18.53
7/8 x 12	1031519	7200	8.83	.88	36.70	24.70	38.20	26.20	2.38	1.25	5.79	12.16
7/8 x 18	1031537	7200	11.5	.88	49.17	31.17	50.67	32.67	2.38	1.25	5.79	18.63
1 x 6	1031555	10000	9.62	1.00	26.24	20.24	28.00	22.00	3.00	1.43	6.50	6.18
1 x 12	1031573	10000	13.0	1.00	38.24	26.24	40.00	28.00	3.00	1.43	6.50	12.18
1 x 18	1031591	10000	16.3	1.00	50.24	32.24	52.00	34.00	3.00	1.43	6.50	18.18
1 x 24	1031617	10000	20.2	1.00	62.84	38.84	64.60	40.60	3.00	1.43	6.47	24.84
1-1/4 x 12	1031635	15200	19.9	1.25	42.14	30.14	44.38	32.38	3.59	1.82	8.49	12.06
1-1/4 x 18	1031653	15200	23.8	1.25	54.14	36.14	56.38	38.38	3.59	1.82	8.49	18.06
1-1/4 x 24	1031671	15200	27.8	1.25	66.70	42.70	68.94	44.94	3.59	1.82	8.49	24.62
1-1/2 x 12	1031699	21400	28.7	1.50	44.24	32.24	46.74	34.74	4.09	2.12	9.46	12.32
1-1/2 x 18	1031715	21400	34.1	1.50	56.24	38.24	58.74	40.74	4.09	2.12	9.46	18.32
1-1/2 x 24	1031733	21400	39.6	1.50	68.86	44.86	71.36	47.36	4.09	2.12	9.46	24.94
1-3/4 x 18	1031779	28000	50.7	1.75	57.38	39.38	60.38	42.38	4.65	2.38	9.97	18.37
1-3/4 x 24	1031797	28000	58.2	1.75	69.38	45.38	72.38	48.38	4.65	2.38	9.97	24.37
2 x 24	1031813	37000	83.5	2.00	75.68	51.68	79.18	55.18	5.81	2.69	13.03	24.48
2-1/2 x 24	1031831	60000	149	2.50	79.18	55.18	83.18	59.18	6.49	3.12	13.76	24.60
2-3/4 x 24	1031859	75000	174	2.75	81.34	57.34	85.84	61.84	7.00	3.25	15.09	24.65

5:1 Design Factor. Proof Load is 2.5 times the Working Load Limit. *Mechanical galvanized

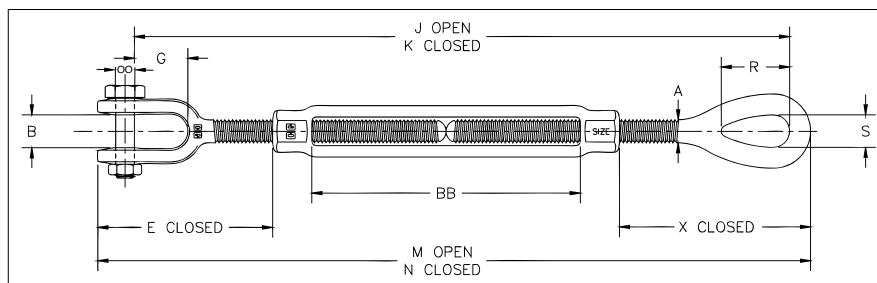


HG-227



APPLICATION AND WARNING INFORMATION
SECTION 17

- End fittings are Quenched & Tempered or normalized, bodies heat-treated by normalizing.
- Hot-dip galvanized steel.
- Turnbuckles eyes are forged and elongated, by design, to maximize easy attachment in system and minimize stress in the eye. For turnbuckles size 1/4" through 2-1/2", a shackle one size smaller can be reeved through eye.
- Forged jaw ends are fitted with bolts and nuts for 1/4" through 5/8", and pins and cotters on 3/4" through 2-3/4" sizes.
- Modified UNJ thread on end fittings for improved fatigue properties.
- Body has UNC threads.
- TURNBUCKLES RECOMMENDED FOR STRAIGHT OR IN-LINE PULL ONLY.
- Lock nuts available for all sizes.
- Fatigue rated to 20,000 cycles at 1-1/2 times the Working Load Limit.
- Meets or exceeds all requirements of ASME B30.26 including identification, ductility, design factor, proof load, and temperature requirements. Importantly, these turnbuckles meet other critical performance requirements including fatigue life, impact properties, and material traceability, not addressed by ASME B30.26.
- Meets the performance requirements of Federal Specifications FF-T-791b, Type 1 Form 1 - CLASS 8, and ASTM F-1145, except for those provisions required of the contractor. For additional information, see warnings and applications section.



HG-227 Jaw & Eye

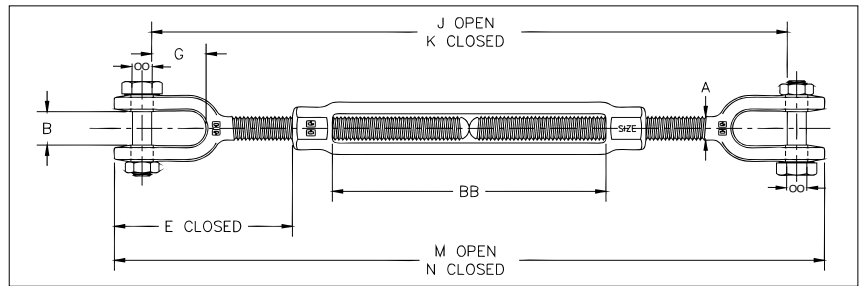
Thread Dia. & Take Up (in)	Stock No.	Working Load Limit (lb)	Weight Each (lb)	Dimensions (in)													Replacement Pin Kit Stock No.
				00	B	E Closed	G	J Open	K Closed	M Open	N Closed	R	S	X Closed	BB		
* 5/16 x 4-1/2	1031895	800	.52	.25	.50	2.02	.87	13.50	9.00	14.30	9.80	.95	.44	2.20	4.58	1089654	
* 3/8 x 6	1031911	1200	.80	.31	.53	2.11	.85	16.91	10.91	17.87	11.87	1.13	.53	2.48	6.10	1089672	
1/2 x 6	1031939	2200	1.77	.37	.64	3.22	1.07	19.30	13.30	20.48	14.48	1.41	.71	3.56	6.03	1089690	
1/2 x 9	1031957	2200	2.25	.37	.64	3.20	1.07	25.59	16.59	26.77	17.77	1.41	.71	3.54	9.36	1089690	
1/2 x 12	1031975	2200	2.67	.37	.64	3.20	1.07	31.59	19.59	32.77	20.77	1.41	.71	3.54	12.36	1089690	
5/8 x 6	1031993	3500	2.98	.50	.79	3.90	1.32	20.73	14.73	22.27	16.27	1.80	.88	4.35	6.03	1089716	
5/8 x 9	1032019	3500	3.72	.50	.79	3.89	1.32	27.07	18.07	28.61	19.61	1.80	.88	4.34	9.39	1089716	
5/8 x 12	1032037	3500	4.35	.50	.79	3.89	1.32	33.07	21.07	34.61	22.61	1.80	.88	4.34	12.39	1089716	
3/4 x 6	1032055	5200	4.51	.63	.97	4.71	1.52	22.17	16.17	24.09	18.09	2.09	1.00	5.12	6.13	1097770	
3/4 x 9	1032073	5200	5.56	.63	.97	4.68	1.52	28.57	19.57	30.49	21.49	2.09	1.00	5.09	9.59	1097770	
3/4 x 12	1032091	5200	6.42	.63	.97	4.68	1.52	34.57	22.57	36.49	24.49	2.09	1.00	5.09	12.59	1097770	
3/4 x 18	1032117	5200	8.14	.63	.97	4.71	1.52	46.57	28.57	48.49	30.49	2.09	1.00	5.12	18.53	1097770	
7/8 x 12	1032135	7200	9.10	.75	1.16	5.50	1.77	35.68	23.68	37.91	25.91	2.38	1.25	5.79	12.16	1097789	
7/8 x 18	1032153	7200	11.6	.75	1.16	5.50	1.77	48.15	30.15	50.38	32.38	2.38	1.25	5.79	18.63	1097789	
1 x 6	1032171	10000	10.0	.88	1.34	6.09	2.05	25.03	19.03	27.59	21.59	3.00	1.43	6.50	6.18	1097761	
1 x 12	1032199	10000	13.4	.88	1.34	6.09	2.05	37.03	25.03	39.59	27.59	3.00	1.43	6.50	12.18	1097761	
1 x 18	1032215	10000	16.7	.88	1.34	6.09	2.05	49.03	31.03	51.59	33.59	3.00	1.43	6.50	18.18	1097761	
1 x 24	1032233	10000	20.6	.88	1.34	6.06	2.05	61.63	37.63	64.19	40.19	3.00	1.43	6.47	24.84	1097761	
1-1/4 x 12	1032251	15200	20.9	1.13	1.84	8.09	2.82	40.76	28.76	43.98	31.98	3.59	1.82	8.49	12.06	1097798	
1-1/4 x 18	1032279	15200	24.8	1.13	1.84	8.09	2.82	52.76	34.76	55.98	37.98	3.59	1.82	8.49	18.06	1097798	
1-1/4 x 24	1032297	15200	28.8	1.13	1.84	8.09	2.82	65.32	41.32	68.54	44.54	3.59	1.82	8.49	24.62	1097798	
1-1/2 x 12	1032313	21400	30.6	1.38	2.06	8.93	2.81	42.50	30.50	46.21	34.21	4.09	2.12	9.46	12.32	1097805	
1-1/2 x 18	1032331	21400	36.0	1.38	2.06	8.93	2.81	54.50	36.50	58.21	40.21	4.09	2.12	9.46	18.32	1097805	
1-1/2 x 24	1032359	21400	41.5	1.38	2.06	8.93	2.81	67.12	43.12	70.83	46.83	4.09	2.12	9.46	24.94	1097805	
1-3/4 x 18	1032395	28000	52.1	1.63	2.60	9.36	3.35	55.37	37.37	59.77	41.77	4.65	2.38	9.97	18.37	1097814	
1-3/4 x 24	1032411	28000	59.7	1.63	2.60	9.36	3.35	67.37	43.37	71.77	47.77	4.65	2.38	9.97	24.37	1097814	
2 x 24	1032439	37000	89.9	2.00	2.62	11.80	3.74	72.66	48.66	77.95	53.95	5.81	2.69	13.03	24.48	1097823	
2-1/2 x 24	1032457	60000	158	2.25	3.06	13.26	4.44	76.08	52.08	82.68	58.68	6.49	3.12	13.76	24.60	1097832	
2-3/4 x 24	1032475	75000	187	2.75	3.69	14.92	4.19	78.05	54.05	85.67	61.67	7.00	3.25	15.09	24.65	1087674	

5:1 Design Factor. Proof Load is 2.5 times the Working Load Limit. *Mechanically galvanized



HG-228

- End fittings are Quenched & Tempered or normalized, bodies heat-treated by normalizing.
- Hot-dip galvanized steel.
- TURNBUCKLES RECOMMENDED FOR STRAIGHT OR IN-LINE PULL ONLY.
- Forged jaw ends are fitted with bolts and nuts for 1/4" through 5/8", and pins and cotters on 3/4" through 2-3/4" sizes.
- Modified UNJ thread on end fittings for improved fatigue properties.
- Body has UNC threads.
- Lock nuts available for all sizes.
- Fatigue rated to 20,000 cycles at 1-1/2 times the Working Load Limit.
- Meets or exceeds all requirements of ASME B30.26 including identification, ductility, design factor, proof load and temperature requirements. Importantly, these turnbuckles meet other critical performance requirements including fatigue life, impact properties and material traceability, not addressed by ASME B30.26.
- Meets the performance requirements of Federal Specifications FF-T-791b, Type 1 Form 1 - CLASS 7, and ASTM F-1145, except for those provisions required of the contractor. For additional information, see Warnings & Applications.



HG-228 Jaw & Jaw

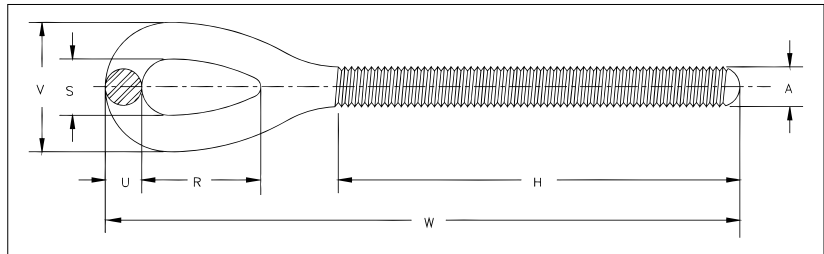
Thread Dia. & Take Up (in)	Stock No.	Working Load Limit (lb)	Weight Each (lb)	Dimensions (in)									Replacement Pin Kit Stock No.
				00	B	E Closed	G	J Open	K Closed	M Open	N Closed	BB	
* 5/16 x 4-1/2	1032518	800	.56	.25	.50	2.02	.87	13.07	8.57	14.12	9.62	4.58	1089654
* 3/8 x 6	1032536	1200	.85	.31	.53	2.11	.85	16.25	10.25	17.50	11.50	6.10	1089672
1/2 x 6	1032554	2200	1.82	.37	.64	3.22	1.07	18.65	12.65	20.14	14.14	6.03	1089690
1/2 x 9	1032572	2200	2.29	.37	.64	3.20	1.07	24.94	15.94	26.43	17.43	9.36	1089690
1/2 x 12	1032590	2200	2.71	.37	.64	3.20	1.07	30.94	18.94	32.43	20.43	12.36	1089690
5/8 x 6	1032616	3500	3.21	.50	.79	3.90	1.32	19.74	13.74	21.82	15.82	6.03	1089716
5/8 x 9	1032634	3500	3.95	.50	.79	3.89	1.32	26.08	17.08	28.16	19.16	9.39	1089716
5/8 x 12	1032652	3500	4.58	.50	.79	3.89	1.32	32.08	20.08	34.16	22.16	12.39	1089716
3/4 x 6	1032670	5200	4.80	.63	.97	4.71	1.52	21.09	15.09	23.68	17.68	6.13	1097770
3/4 x 9	1032698	5200	5.85	.63	.97	4.68	1.52	27.49	18.49	30.08	21.08	9.59	1097770
3/4 x 12	1032714	5200	6.72	.63	.97	4.68	1.52	33.49	21.49	36.08	24.08	12.59	1097770
3/4 x 18	1032732	5200	8.45	.63	.97	4.71	1.52	45.49	27.49	48.08	30.08	18.53	1097770
7/8 x 12	1032750	7200	9.37	.75	1.16	5.50	1.77	34.65	22.65	37.62	25.62	12.16	1097789
7/8 x 18	1032778	7200	11.8	.75	1.16	5.50	1.77	47.12	29.12	50.09	32.09	18.63	1097789
1 x 6	1032796	10000	10.4	.88	1.34	6.09	2.05	23.82	17.82	27.18	21.18	6.18	1097761
1 x 12	1032812	10000	13.8	.88	1.34	6.09	2.05	35.82	23.82	39.18	27.18	12.18	1097761
1 x 18	1032830	10000	17.1	.88	1.34	6.09	2.05	47.82	29.82	51.18	33.18	18.18	1097761
1 x 24	1032858	10000	21.0	.88	1.34	6.06	2.05	60.42	36.42	63.78	39.78	24.84	1097761
1-1/4 x 12	1032876	15200	21.9	1.13	1.84	8.09	2.82	39.37	27.37	43.58	31.58	12.06	1097798
1-1/4 x 18	1032894	15200	25.9	1.13	1.84	8.09	2.82	51.37	33.37	55.58	37.58	18.06	1097798
1-1/4 x 24	1032910	15200	29.8	1.13	1.84	8.09	2.82	63.93	39.93	68.14	44.14	24.62	1097798
1-1/2 x 12	1032938	21400	32.6	1.38	2.06	8.93	2.81	40.76	28.76	45.68	33.68	12.32	1097805
1-1/2 x 18	1032956	21400	38.0	1.38	2.06	8.93	2.81	52.76	34.76	57.68	39.68	18.32	1097805
1-1/2 x 24	1032974	21400	43.5	1.38	2.06	8.93	2.81	65.38	41.38	70.30	46.30	24.94	1097805
1-3/4 x 18	1033018	28000	53.5	1.63	2.60	9.36	3.35	53.35	35.35	59.16	41.16	18.37	1097814
1-3/4 x 24	1033036	28000	61.1	1.63	2.60	9.36	3.35	65.35	41.35	71.16	47.16	24.37	1097814
2 x 24	1033054	37000	96.3	2.00	2.62	11.80	3.74	69.64	45.64	76.72	52.72	24.48	1097823
2-1/2 x 24	1033072	60000	167	2.25	3.06	13.26	4.44	72.97	48.97	82.18	58.18	24.60	1097832
2-3/4 x 24	1033090	75000	199	2.75	3.69	14.92	4.19	74.75	50.75	85.50	61.50	24.65	1087674

5:1 Design Factor. Proof Load is 2.5 times the Working Load Limit. *Mechanical galvanized



HG-4037

- Quenched & Tempered or normalized.
- Hot-dip galvanized steel.
- Turnbuckle eyes are forged elongated, by design, to maximize easy attachment in system and minimize stress in the eye. For turnbuckle sizes 1/4" through 2-1/2", a shackle one size smaller can be reeved through eye.
- Modified UNJ thread for improved fatigue properties.
- Fatigue rated.



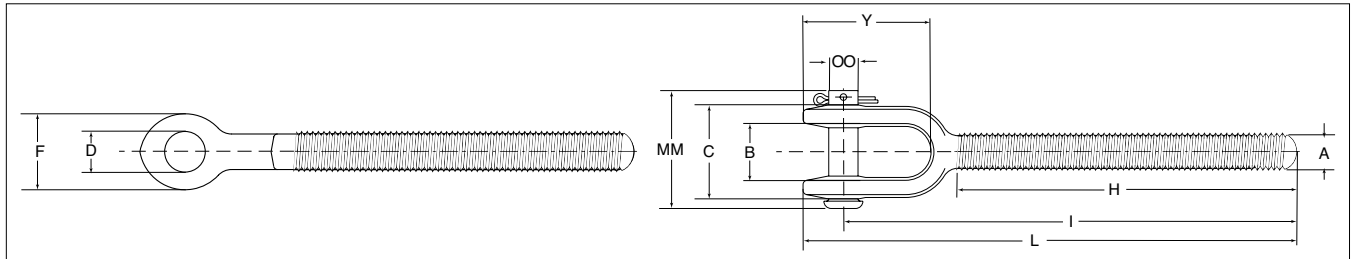
HG-4037 Eye End Fittings

Shank Dia. & Take Up (in)	RH Eye Stock No.	LH Eye Stock No.	Working Load Limit (lb)	Weight Each (lb)	Dimensions (in)						
					A	H	R	S	U	V	W
* 1/4 x 4	1071057	1071672	500	.07	.25	2.59	.81	.34	.22	.78	4.19
* 5/16 x 4 1/2	1071075	1071690	800	.13	.31	3.00	.95	.44	.28	1.00	4.99
* 3/8 x 6	1071093	1071716	1200	.23	.38	3.88	1.13	.53	.34	1.21	6.12
1/2 x 6	1071119	1071734	2200	.51	.50	4.19	1.41	.71	.44	1.59	7.41
1/2 x 9	1071137	1071752	2200	.59	.50	5.69	1.41	.71	.44	1.59	8.91
1/2 x 12	1071155	1071770	2200	.68	.50	7.19	1.41	.71	.44	1.59	10.41
5/8 x 6	1071173	1071798	3500	.82	.63	4.44	1.80	.88	.50	1.88	8.36
5/8 x 9	1071191	1071814	3500	.95	.63	5.94	1.80	.88	.50	1.88	9.86
5/8 x 12	1071217	1071832	3500	1.08	.63	7.44	1.80	.88	.50	1.88	11.36
3/4 x 6	1071235	1071850	5200	1.36	.75	4.56	2.09	1.00	.63	2.26	9.25
3/4 x 9	1071253	1071878	5200	1.55	.75	6.06	2.09	1.00	.63	2.26	10.75
3/4 x 12	1071271	1071896	5200	1.73	.75	7.56	2.09	1.00	.63	2.26	12.25
3/4 x 18	1071299	1071912	5200	2.10	.75	10.56	2.09	1.00	.63	2.26	15.25
7/8 x 12	1071315	1071930	7200	2.61	.88	7.81	2.38	1.25	.75	2.75	13.10
7/8 x 18	1071333	1071958	7200	3.12	.88	10.81	2.38	1.25	.75	2.75	16.10
1 x 6	1071351	1071976	10000	3.15	1.00	5.06	3.00	1.43	.88	3.19	11.00
1 x 12	1071379	1071994	10000	3.81	1.00	8.06	3.00	1.43	.88	3.19	14.00
1 x 18	1071397	1072010	10000	4.48	1.00	11.06	3.00	1.43	.88	3.19	17.00
1 x 24	1071413	1072038	10000	5.15	1.00	14.06	3.00	1.43	.88	3.19	20.00
1-1/4 x 12	1071431	1072056	15200	7.07	1.25	8.38	3.59	1.82	1.12	4.06	16.19
1-1/4 x 18	1071459	1072074	15200	8.12	1.25	11.38	3.59	1.82	1.12	4.06	19.19
1-1/4 x 24	1071477	1072092	15200	9.16	1.25	14.38	3.59	1.82	1.12	4.06	22.19
1-1/2 x 12	1071495	1072118	21400	10.3	1.50	8.75	4.09	2.12	1.25	4.62	17.37
1-1/2 x 18	1071510	1072136	21400	11.8	1.50	11.75	4.09	2.12	1.25	4.62	20.37
1-1/2 x 24	1071538	1072154	21400	13.3	1.50	14.75	4.09	2.12	1.25	4.62	23.37
1-3/4 x 18	1071574	1072190	28000	17.5	1.75	12.16	4.65	2.38	1.50	5.38	21.19
1-3/4 x 24	1071592	1072216	28000	19.5	1.75	15.16	4.65	2.38	1.50	5.38	24.19
2 x 24	1071618	1072234	37000	28.9	2.00	15.59	5.81	2.69	1.75	6.19	27.59
2-1/2 x 24	1071636	1072252	60000	46.4	2.50	17.56	6.50	3.12	2.00	7.12	29.59
2-3/4 x 24	1071654	1072270	75000	60.2	2.75	17.69	7.00	3.25	2.25	7.75	30.92

*Mechanically galvanized

HG-4037 Jaw End Fittings

- Quenched & Tempered or normalized.
- Hot-dip galvanized steel.
- Forged jaw ends are fitted with bolts and nuts on sizes 1/4" through 5/8", and pins and cotters on sizes 3/4" through 2-3/4".
- Modified UNJ thread for improved fatigue properties.
- Fatigue Rated.



QT Fatigue Rated

HG-4037 Jaw End Fittings

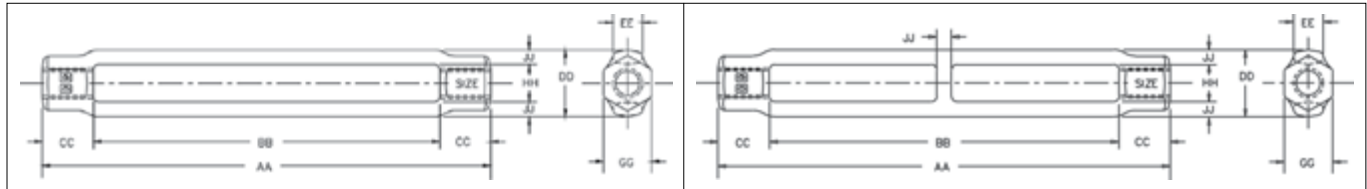
Shank Dia. & Take Up (in)	RH Jaw Stock No.	LH Jaw Stock No.	Working Load Limit (lb)	Weight Each (lb)	Dimensions (in)										
					A	B	C	D	F	H	I Nom. Min.	L Nom. Min.	Y	MM	OO Bolt Pin
* 1/4 x 4	1072298	1072911	500	.11	.25	.45	.91	.30	.63	2.59	3.72	4.09	1.13	1.41	.25
* 5/16 x 4 1/2	1072314	1072939	800	.17	.31	.50	1.02	.30	.69	3.00	4.41	4.81	1.39	1.41	.25
* 3/8 x 6	1072332	1072957	1200	.28	.38	.53	1.15	.36	.81	3.88	5.28	5.75	1.47	1.58	.31
1/2 x 6	1072350	1072975	2200	.56	.50	.64	1.36	.42	1.00	4.19	6.51	7.07	1.81	1.87	.37
1/2 x 9	1072378	1072993	2200	.63	.50	.64	1.36	.42	1.00	5.69	8.01	8.57	1.81	1.87	.37
1/2 x 12	1072396	1073019	2200	.72	.50	.64	1.36	.42	1.00	7.19	9.51	10.07	1.81	1.87	.37
5/8 x 6	1072412	1073037	3500	1.05	.63	.79	1.75	.55	1.31	4.31	7.12	7.91	2.36	2.44	.50
5/8 x 9	1072430	1073055	3500	1.18	.63	.79	1.75	.55	1.31	5.81	8.62	9.41	2.36	2.44	.50
5/8 x 12	1072458	1073073	3500	1.31	.63	.79	1.75	.55	1.31	7.31	10.12	10.91	2.36	2.44	.50
3/4 x 6	1072476	1073091	5200	1.65	.75	.97	2.09	.69	1.63	4.56	7.86	8.84	2.81	2.56	.63
3/4 x 9	1072494	1073117	5200	1.84	.75	.97	2.09	.69	1.63	6.06	9.36	10.34	2.81	2.56	.63
3/4 x 12	1072519	1073135	5200	2.03	.75	.97	2.09	.69	1.63	7.56	10.86	11.84	2.81	2.56	.63
3/4 x 18	1072537	1073153	5200	2.41	.75	.97	2.09	.69	1.63	10.56	13.86	14.84	2.81	2.56	.63
7/8 x 12	1072555	1073171	7200	2.88	.88	1.16	2.56	.81	1.88	7.81	11.70	12.81	3.25	3.09	.75
7/8 x 18	1072573	1073199	7200	3.25	.88	1.16	2.56	.81	1.88	10.81	14.70	15.81	3.25	3.09	.75
1 x 6	1072591	1073215	10000	3.56	1.00	1.34	2.76	.94	2.12	5.06	9.35	10.59	3.73	3.44	.88
1 x 12	1072617	1073233	10000	4.22	1.00	1.34	2.76	.94	2.12	8.06	12.35	13.59	3.73	3.44	.88
1 x 18	1072635	1073251	10000	4.89	1.00	1.34	2.76	.94	2.12	11.06	15.35	16.59	3.73	3.44	.88
1 x 24	1072653	1073279	10000	5.56	1.00	1.34	2.76	.94	2.12	14.06	18.35	19.59	3.73	3.44	.88
1-1/4 x 12	1072671	1073297	15200	8.10	1.25	1.84	3.72	1.19	2.63	8.38	14.25	15.79	4.92	4.53	1.13
1-1/4 x 18	1072699	1073313	15200	9.14	1.25	1.84	3.72	1.19	2.63	11.38	17.25	18.79	4.92	4.53	1.13
1-1/4 x 24	1072715	1073331	15200	10.2	1.25	1.84	3.72	1.19	2.63	14.38	20.25	21.79	4.92	4.53	1.13
1-1/2 x 12	1072733	1073359	21400	12.3	1.50	2.06	4.16	1.47	3.12	8.75	15.07	16.84	5.27	5.13	1.38
1-1/2 x 18	1072751	1073377	21400	13.8	1.50	2.06	4.16	1.47	3.12	11.75	18.07	19.84	5.27	5.13	1.38
1-1/2 x 24	1072779	1073395	21400	15.3	1.50	2.06	4.16	1.47	3.12	14.75	21.07	22.84	5.27	5.13	1.38
1-3/4 x 18	1072813	1073439	28000	18.9	1.75	2.60	4.66	1.72	3.50	12.16	18.49	20.58	6.25	6.00	1.63
1-3/4 x 24	1072831	1073457	28000	21.0	1.75	2.60	4.66	1.72	3.50	15.16	21.49	23.58	6.25	6.00	1.63
2 x 24	1072859	1073475	37000	35.3	2.00	2.62	5.61	2.09	4.19	15.59	23.82	26.36	7.28	6.88	2.00
2-1/2 x 24	1072877	1073493	60000	55.8	2.50	3.06	5.84	2.38	5.62	17.20	25.61	29.09	9.04	7.50	2.25
2-3/4 x 24	1072895	1073518	75000	72.4	2.75	3.69	6.57	2.88	6.12	17.35	26.75	30.75	9.56	8.38	2.75

*Mechanical galvanized

10

HG-2510 Body

- Heat treat by normalizing.
- Hot-dip galvanized
- UNC threads.
- Fatigue rated.
- Meets the performance requirements of Federal Specifications FF-T-791b, Type 1, Form 1 - Class 2, except for those provisions required by the contractor.



HG-2510 Body



Shank Dia. & Take Up (in)	Stock No.	Working Load Limit (lb)	Weight Each (lb)	Dimensions (in)							
				AA	BB	CC	DD	EE	GG	HH	JJ
* 5/16 x 4-1/2	1033919	800	.22	5.59	4.58	.51	.82	.38	.56	.44	.19
* 3/8 x 6	1033937	1200	.29	7.29	6.10	.60	.88	.38	.63	.50	.19
1/2 x 6	1033955	2200	.70	7.70	6.03	.84	1.19	.68	.81	.63	.28
† 1/2 x 9	1033973	2200	1.03	11.03	9.36	.84	1.19	.68	.81	.63	.28
† 1/2 x 12	1033991	2200	1.27	14.03	12.36	.84	1.19	.68	.81	.63	.28
5/8 x 6	1034017	3500	1.11	8.02	6.03	1.00	1.43	.83	1.00	.75	.34
† 5/8 x 9	1034035	3500	1.59	11.38	9.39	1.00	1.43	.83	1.00	.75	.34
† 5/8 x 12	1034053	3500	1.96	14.38	12.39	1.00	1.43	.83	1.00	.75	.34
3/4 x 6	1034071	5200	1.50	8.26	6.13	1.07	1.74	.94	1.13	.94	.40
† 3/4 x 9	1034099	5200	2.17	11.72	9.59	1.07	1.74	.94	1.13	.94	.40
† 3/4 x 12	1034115	5200	2.66	14.72	12.59	1.07	1.74	.94	1.13	.94	.40
† 3/4 x 18	1034133	5200	3.63	20.66	18.53	1.07	1.74	.94	1.13	.94	.40
7/8 x 12	1034179	7200	3.61	14.62	12.16	1.23	2.00	1.13	1.31	1.06	.47
† 7/8 x 18	1034197	7200	5.27	21.09	18.63	1.23	2.00	1.13	1.31	1.06	.47
1 x 6	1034213	10000	3.32	9.00	6.18	1.41	2.45	1.25	1.50	1.25	.60
1 x 12	1034231	10000	5.34	15.00	12.18	1.41	2.45	1.25	1.50	1.25	.60
† 1 x 18	1034259	10000	7.35	21.00	18.18	1.41	2.45	1.25	1.50	1.25	.60
† 1 x 24	1034277	10000	9.85	27.66	24.84	1.41	2.45	1.25	1.50	1.25	.60
1-1/4 x 12	1034339	15200	5.72	15.40	12.06	1.67	2.62	1.25	1.88	1.50	.56
1-1/4 x 18	1034357	15200	7.58	21.40	18.06	1.67	2.62	1.25	1.88	1.50	.56
† 1-1/4 x 24	1034375	15200	9.45	27.96	24.62	1.67	2.62	1.25	1.88	1.50	.56
1-1/2 x 12	1034437	21400	8.01	15.82	12.32	1.75	2.99	1.50	2.25	1.75	.62
1-1/2 x 18	1034455	21400	10.4	21.82	18.32	1.75	2.99	1.50	2.25	1.75	.62
† 1-1/2 x 24	1034473	21400	12.9	28.45	24.94	1.75	2.99	1.50	2.25	1.75	.62
1-3/4 x 18	1034552	28000	15.7	22.44	18.37	2.04	3.62	1.75	2.62	2.12	.75
1-3/4 x 24	1034570	28000	19.2	28.44	24.37	2.04	3.62	1.75	2.62	2.12	.75
2 x 24	1034632	37000	25.8	29.13	24.48	2.33	4.14	2.00	3.00	2.38	.88
2-1/2 x 24	1034678	60000	55.9	31.66	24.60	3.53	5.62	2.75	3.88	3.12	1.25
2-3/4 x 24	1034696	75000	54.0	31.66	24.65	3.51	5.62	2.75	3.88	4.48	1.25

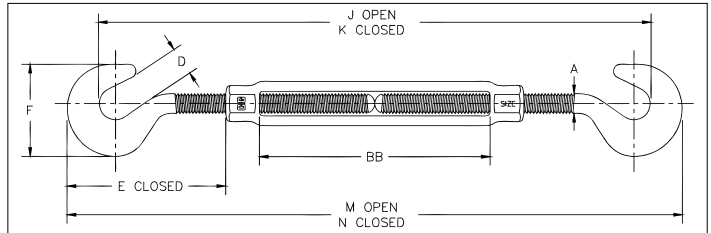
*Mechanically galvanized

†Contains Center Rib for additional body support



HG-223

- End fittings are Quenched & Tempered or normalized, bodies heat-treated by normalizing.
- Hot-dip galvanized steel.
- Hooks are forged with a greater cross sectional area that results in a stronger hook with better fatigue properties.
- TURNBUCKLES RECOMMENDED FOR STRAIGHT OR IN-LINE PULL ONLY.
- Modified UNJ thread on end fittings for improved fatigue properties.
- Body has UNC threads.
- Lock nuts available for all sizes.
- Fatigue rated to 20,000 cycles at 1-1/2 times the Working Load Limit.
- Meets or exceeds all requirements of ASME B30.26 including identification, ductility, design factor, proof load and temperature requirements. Importantly, these turnbuckles meet other critical performance requirements including fatigue life, impact properties and material traceability, not addressed by ASME B30.26.
- Meets the performance requirements of Federal Specifications FF-T-791b, Type 1 Form 1 - CLASS 5, and ASTM F-1145, except for those provisions required of the contractor. For additional information, see warnings and applications section.



HG-223 Hook & Hook

Thread Dia. & Take Up (in)	Stock No.	Working Load Limit (lb)*	Weight Each (lb)	Dimensions (in)								
				A	D	E Closed	F	J Open	K Closed	M Open	N Closed	BB
1 x 12	1030333	5000	14.8	1.00	1.25	6.56	4.25	36.59	25.06	40.12	28.12	12.18

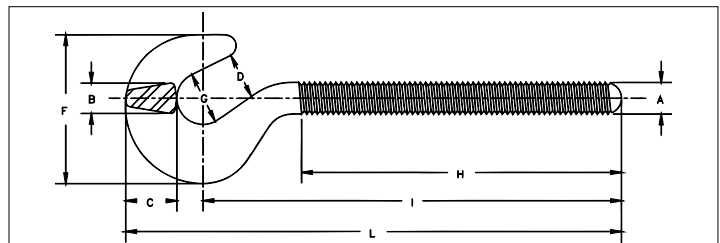
5:1 Design Factor. Proof Load is 2.5 times the Working Load Limit.

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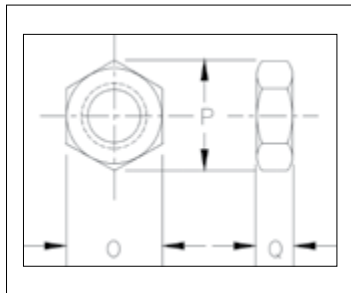
HG-4037

- Quenched & Tempered or normalized.
- Hot-dip galvanized steel.
- Hooks are forged with a greater cross sectional area that results in a stronger hook with better fatigue properties.
- Fatigue rated to 20,000 cycles at 1-1/2 times the Working Load Limit.



HG-4037 Hook End Fittings

Shank Dia. & Take Up (in)	RH Hook Stock No.	LH Hook Stock No.	Working Load Limit (lb)	Weight Each (lb)	Dimensions (in)								
					A	B	C	D	F	G	H	I	L
1 x 12	1070334	1070851	5000	4.72	1.00	1.00	1.53	1.25	4.25	1.38	8.06	11.84	14.06

**HG-4060 / HG-4061**

- Secures the turnbuckle into position at final adjustment.

HG-4060 / HG-4061 Lock Nuts

Shank Dia. & Take Up (in)	Right Hand HG-4060 Stock No.	Left Hand HG-4061 Stock No.	Weight Per 100 (lb)	Dimensions (in)		
				O	P	Q
1/4	1075115	1075491	.80	.44	.50	.16
5/16	1075133	1075516	1.30	.50	.56	.19
3/8	1075151	1075534	2.00	.56	.64	.22
1/2	1075197	1075570	4.00	.75	.86	.31
5/8	1075213	1075598	7.00	.94	1.06	.38
3/4	1075231	1075614	11.00	1.13	1.26	.42
7/8	1075259	1075632	16.30	1.31	1.50	.48
1	1075277	1075650	23.80	1.50	1.69	.55
1-1/8	1075295	1075678	32.00	1.50	1.69	.55
1-1/4	1075311	1075696	62.50	1.88	2.13	.72
1-1/2	1075357	1075730	72.00	2.25	2.53	.84
1-3/4	1075393	1075776	112.00	2.75	3.18	1.00
2	1075419	1075794	150.00	3.12	3.61	1.12
2-1/2	1075455	1075838	330.00	3.88	4.47	1.50
2-3/4	1075473	1075856	425.00	4.25	4.91	1.62

GREEN PIN[®]

PRODUCT CATALOGUE

IMPERIAL



TURNBUCKLES



Applications

Turnbuckles are used for rigging or tensioning wires, ropes, rods etc. They are designed for in-line rigging, tensioning or lashing. Green Pin® Turnbuckles (G-6313, G-6323, G-6333, G-6311, G-6312, G-6315 and G-6314) can be used in lifting applications. The closed body rigging screws (G-6343, G-6340 and G-6345) can also be used in lifting applications.

Range

Green Pin® offers a wide range of turnbuckles:

- Load rated Green Pin® turnbuckles;
- Open body rigging screws generally to DIN 1480;
- Rigging screws with welding ends;
- Closed body rigging screws;
- Special turnbuckles for lashing (hamburgers).

Royal Van Beest offers a wide range of other turnbuckles to complement the Green Pin® assortment.

Design

Green Pin® turnbuckles are manufactured to ASTM F1145-92 (formerly U.S. Fed. Spec. FF-T-791). They are drop-forged and available with the following end fittings: eye/eye, hook/hook, hook/eye, jaw/jaw and jaw/eye. All fittings are interchangeable. Locking nuts are supplied with all sizes.

All Green Pin® turnbuckles are generally marked with:

- | | |
|-------------------------|--------------------------------------|
| • Working Load Limit | - e.g. 2.36 t |
| • manufacturer's symbol | - e.g. GP |
| • thread diameter | - e.g. 3/4" |
| • traceability code | - e.g. A1 |
| • thread | - L (lefthanded) and R (righthanded) |

Rigging screws generally to DIN 1480 are available with welding ends and in hook/eye, eye/eye, hook/hook and jaw/jaw combinations. Closed body rigging screws are available in jaw/jaw, jaw/eye and eye/eye combinations.

Finish

Load rated Green Pin® turnbuckles and closed body rigging screws are hot dipped galvanized. Rigging screws to DIN 1480 are electro-galvanized. Lashing turnbuckles are self-coloured.

Certification

Specific details of certificate availability can be found on each product page. Please verify your certification requirements at the time of order.

Instructions for use

Turnbuckles must be used for in-line applications only. Special attention should be paid to prevent overloading. During tensioning, avoid forces on the turnbuckle that may lead to deformation. Should a turnbuckle start to deform, the tension should be decreased immediately and any deformed parts should be replaced. Should extreme circumstances or shock loading, possibly occur during use, this must be taken into account when selecting the correct products to be used for the application.

For the rigging of wires, ropes, rods etc., Green Pin® turnbuckles are recommended to be used. The WLL should be applied in in-line lifting only and overloading is not permitted. Nor should side loads be applied, as the products have not been designed for this purpose.

Open body rigging screws are used for tensioning wires and ropes for less demanding applications (for example rope railings).

Turnbuckles must be regularly inspected in accordance with the safety standards and regulations given in the country of use. This is required because the products in use may be affected by wear, misuse, overloading etc. which may lead to deformation and alteration of the steel structure.

Safe use of turnbuckles

Turnbuckles should be inspected before use to ensure that:

- all markings are legible;
- the threads of the body and the end fittings are of the same type;
- the pin, nut, cotter pin or any other locking system cannot vibrate out of position;
- the threads of the body and the end fittings are undamaged;
- the body and end fittings are not distorted or unduly worn;
- the body and end fittings are free from nicks, gouges and cracks.

Make sure that the end fittings are correctly screwed into the body. Always use the locking nuts provided to prevent the turnbuckles from unscrewing. Never replace an end fitting by anything other than one designed for the purpose, otherwise the turnbuckle may not be suitable for the loads imposed.



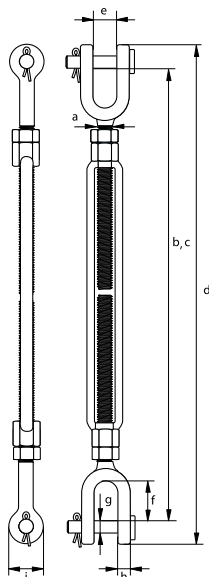
Green Pin® JJ Turnbuckle CP

Turnbuckle with jaw-jaw end-fitting and cotter pins, generally to ASTM F1145-92

Scan for additional product details



G-6313



- **Material:** drop forged high tensile steel SAE 1035 or 1045
- **Safety factor:** MBL equals 5 x WLL
- **Standard:** generally to ASTM F1145-92
formerly U.S. Federal Specification FF-T-791b
hot dipped galvanized
- **Finish:** hot dipped galvanized
- **Temperature range:** -4 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 MTC^a CE IIA
- **Article code:** scan QR code to see article codes

working load limit	diameter thread		take up	length closed position	length open position	length closed position	opening jaw	length inside	diameter pin	thickness jaw eye	diameter jaw eye	weight each
metric t	a inch	a mm	inch	b inch	c inch	d inch	e inch	f inch	g inch	h inch	i inch	lbs
2.36	3/4	19	6	14 17/32	19 3/16	17 9/32	15/16	1 1/2	5/8	5/8	1 5/8	5.71
2.36	3/4	19	9	17 15/32	25 3/16	20 1/4	15/16	1 1/2	5/8	5/8	1 5/8	6.9
2.36	3/4	19	12	20 15/32	31 3/16	23 7/32	15/16	1 1/2	5/8	5/8	1 5/8	7.54
2.36	3/4	19	18	26 3/8	43 5/32	29 1/8	15/16	1 1/2	5/8	5/8	1 5/8	9.94
3.27	7/8	22	12	22 3/32	32 17/32	25 1/8	1 1/16	1 21/32	3/4	3/4	1 7/8	10.87
3.27	7/8	22	18	28 1/16	44 9/16	31 3/32	1 1/16	1 21/32	3/4	3/4	1 7/8	14.13
4.54	1	25	6	17 19/32	21 13/16	20 15/16	1 7/32	1 31/32	7/8	25/32	2 1/8	11.42
4.54	1	25	12	23 17/32	33 13/16	26 7/8	1 7/32	1 31/32	7/8	25/32	2 1/8	14.18
4.54	1	25	18	29 17/32	45 31/32	32 7/8	1 7/32	1 31/32	7/8	25/32	2 1/8	17.81
4.5	1	25	24	35 9/16	57 7/8	38 29/32	1 7/32	1 31/32	7/8	25/32	2 1/8	18.87
6.9	1 1/4	32	12	25 1/4	35 31/32	29 7/16	1 23/32	2 25/32	1 5/32	1 1/32	2 11/16	24.7
6.9	1 1/4	32	18	31 5/8	48 11/32	35 13/16	1 23/32	2 25/32	1 5/32	1 1/32	2 11/16	30
6.9	1 1/4	32	24	37 7/8	60 19/32	42 3/32	1 23/32	2 25/32	1 5/32	1 1/32	2 11/16	33.1
9.71	1 1/2	38	12	26 9/16	37 3/32	31 23/32	2 1/16	2 25/32	1 3/8	1 3/32	3 5/32	37.7
9.71	1 1/2	38	18	32 15/32	48 31/32	37 5/8	2 1/16	2 25/32	1 3/8	1 3/32	3 5/32	42.5
9.71	1 1/2	38	24	38 19/32	61 1/16	43 3/4	2 1/16	2 25/32	1 3/8	1 3/32	3 5/32	48.1
12.7	1 3/4	45	18	36 15/16	51 15/16	43	2 3/8	3 3/8	1 5/8	1 5/16	3 17/32	60
12.7	1 3/4	45	24	42 7/8	63 13/16	48 15/16	2 3/8	3 3/8	1 5/8	1 5/16	3 17/32	70
16.8	2	50	24	45 5/16	65 25/32	52 11/16	2 15/32	3 21/32	2	1 9/16	4 7/32	108
27.2	2 1/2	64	24	49 13/32	72 3/32	58 9/32	2 15/16	4 1/2	2 1/4	1 5/8	5 5/8	194
34	2 3/4	69	24	53 1/16	74 3/32	63 5/32	3 17/32	4 11/32	2 3/4	1 5/8	6 7/32	227

CAD INFO

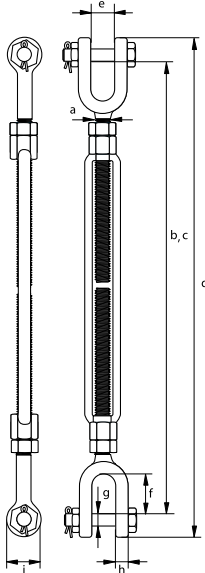
Green Pin® JJ Turnbuckle BN

Turnbuckle with jaw-jaw end-fitting and safety bolt, generally to ASTM F1145-92

Scan for additional product details



G-6323



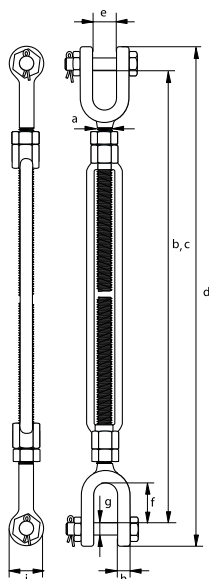
- **Material:** drop forged high tensile steel SAE 1035 or 1045
- **Safety factor:** MBL equals 5 x WLL
- **Standard:** generally to ASTM F1145-92
formerly U.S. Federal Specification FF-T-791b
hot dipped galvanized
- **Finish:**
- **Temperature range:** -4 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 MTC³ CE IIA
- **Article code:** scan QR code to see article codes

working load limit	diameter thread		take up	length closed position	length open position	length closed position	opening jaw	length inside	diameter pin	thickness jaw eye	diameter jaw eye	weight each
metric t	a inch	a mm	inch	b inch	c inch	d inch	e inch	f inch	g inch	h inch	i inch	lbs
0.54	³ / ₈	10	6	10 ³ / ₄	16 ³ / ₃₂	11 ³¹ / ₃₂	¹⁵ / ₃₂	¹³ / ₁₆	⁹ / ₃₂	¹¹ / ₃₂	¹³ / ₁₆	1.21
1	¹ / ₂	12	6	11 ³¹ / ₃₂	17 ¹ / ₈	13 ¹ / ₂	⁵ / ₈	1 ¹ / ₃₂	¹³ / ₃₂	⁷ / ₁₆	³¹ / ₃₂	2.14
1	¹ / ₂	12	9	14 ²⁹ / ₃₂	23 ⁵ / ₃₂	16 ¹⁵ / ₃₂	⁵ / ₈	1 ¹ / ₃₂	¹³ / ₃₂	⁷ / ₁₆	³¹ / ₃₂	2.54
1	¹ / ₂	12	12	17 ²⁹ / ₃₂	29 ⁷ / ₈	19 ⁷ / ₁₆	⁵ / ₈	1 ¹ / ₃₂	¹³ / ₃₂	⁷ / ₁₆	³¹ / ₃₂	3.22
1.59	⁵ / ₈	16	6	13 ⁵ / ₈	18 ¹⁵ / ₃₂	15 ³¹ / ₃₂	²³ / ₃₂	1 ¹ / ₄	¹ / ₂	⁹ / ₁₆	1 ⁵ / ₁₆	3.86
1.59	⁵ / ₈	16	9	16 ⁹ / ₁₆	24 ¹ / ₂	18 ²⁹ / ₃₂	²³ / ₃₂	1 ¹ / ₄	¹ / ₂	⁹ / ₁₆	1 ⁵ / ₁₆	4.72
1.59	⁵ / ₈	16	12	19 ¹⁹ / ₃₂	30 ¹⁵ / ₃₂	21 ¹⁵ / ₁₆	²³ / ₃₂	1 ¹ / ₄	¹ / ₂	⁹ / ₁₆	1 ⁵ / ₁₆	5.36
2.36	³ / ₄	19	6	14 ¹⁷ / ₃₂	19 ³ / ₁₆	17 ⁹ / ₃₂	¹⁵ / ₁₆	1 ¹ / ₂	⁵ / ₈	⁵ / ₈	1 ⁵ / ₈	5.95
2.36	³ / ₄	19	9	17 ¹⁵ / ₃₂	25 ³ / ₁₆	20 ¹ / ₄	¹⁵ / ₁₆	1 ¹ / ₂	⁵ / ₈	⁵ / ₈	1 ⁵ / ₈	7.12
2.36	³ / ₄	19	12	20 ¹⁵ / ₃₂	31 ³ / ₁₆	23 ⁷ / ₃₂	¹⁵ / ₁₆	1 ¹ / ₂	⁵ / ₈	⁵ / ₈	1 ⁵ / ₈	7.87
2.36	³ / ₄	19	18	26 ³ / ₈	43 ⁵ / ₃₂	29 ¹ / ₈	¹⁵ / ₁₆	1 ¹ / ₂	⁵ / ₈	⁵ / ₈	1 ⁵ / ₈	10.03
3.27	⁷ / ₈	22	12	22 ³ / ₃₂	32 ¹⁷ / ₃₂	25 ¹ / ₈	1 ¹ / ₁₆	1 ²¹ / ₃₂	³ / ₄	³ / ₄	1 ⁷ / ₈	11.51
3.27	⁷ / ₈	22	18	28 ¹ / ₁₆	44 ⁹ / ₁₆	31 ³ / ₃₂	1 ¹ / ₁₆	1 ²¹ / ₃₂	³ / ₄	³ / ₄	1 ⁷ / ₈	14.46
4.54	1	25	6	17 ¹⁹ / ₃₂	21 ¹³ / ₁₆	20 ¹⁵ / ₁₆	1 ⁷ / ₃₂	1 ³¹ / ₃₂	⁷ / ₈	²⁵ / ₃₂	2 ¹ / ₈	12.21
4.54	1	25	12	23 ¹⁷ / ₃₂	33 ¹³ / ₁₆	26 ⁷ / ₈	1 ⁷ / ₃₂	1 ³¹ / ₃₂	⁷ / ₈	²⁵ / ₃₂	2 ¹ / ₈	14.77
4.54	1	25	18	29 ¹⁷ / ₃₂	45 ³¹ / ₃₂	32 ⁷ / ₈	1 ⁷ / ₃₂	1 ³¹ / ₃₂	⁷ / ₈	²⁵ / ₃₂	2 ¹ / ₈	18.98
4.5	1	25	24	35 ⁹ / ₁₆	57 ⁷ / ₈	38 ²⁹ / ₃₂	1 ⁷ / ₃₂	1 ³¹ / ₃₂	⁷ / ₈	²⁵ / ₃₂	2 ¹ / ₈	19.56
6.9	1 ¹ / ₄	32	12	25 ⁵ / ₁₆	36 ¹ / ₁₆	29 ⁷ / ₁₆	1 ²³ / ₃₂	2 ²⁵ / ₃₂	1 ³ / ₃₂	1 ¹ / ₃₂	2 ¹¹ / ₁₆	26.2
6.9	1 ¹ / ₄	32	18	31 ¹¹ / ₁₆	48 ⁷ / ₁₆	35 ¹³ / ₁₆	1 ²³ / ₃₂	2 ²⁵ / ₃₂	1 ³ / ₃₂	1 ¹ / ₃₂	2 ¹¹ / ₁₆	30
6.9	1 ¹ / ₄	32	24	37 ¹⁵ / ₁₆	60 ²¹ / ₃₂	42 ³ / ₃₂	1 ²³ / ₃₂	2 ²⁵ / ₃₂	1 ³ / ₃₂	1 ¹ / ₃₂	2 ¹¹ / ₁₆	31.3
9.71	1 ¹ / ₂	38	12	26 ⁹ / ₁₆	37 ³ / ₃₂	31 ²³ / ₃₂	2 ¹ / ₁₆	2 ²⁵ / ₃₂	1 ³ / ₈	1 ³ / ₃₂	3 ⁵ / ₃₂	40.8
9.71	1 ¹ / ₂	38	18	32 ¹⁵ / ₃₂	48 ³¹ / ₃₂	37 ⁵ / ₈	2 ¹ / ₁₆	2 ²⁵ / ₃₂	1 ³ / ₈	1 ³ / ₃₂	3 ⁵ / ₃₂	42.5
9.71	1 ¹ / ₂	38	24	38 ¹⁹ / ₃₂	61 ¹ / ₁₆	43 ³ / ₄	2 ¹ / ₁₆	2 ²⁵ / ₃₂	1 ³ / ₈	1 ³ / ₃₂	3 ⁵ / ₃₂	48.5
12.7	1 ³ / ₄	45	18	36 ¹⁵ / ₁₆	51 ¹³ / ₁₆	43	2 ³ / ₈	3 ³ / ₈	1 ⁵ / ₈	1 ⁵ / ₁₆	3 ¹⁷ / ₃₂	64
12.7	1 ³ / ₄	45	24	42 ⁷ / ₈	63 ¹³ / ₁₆	48 ¹⁵ / ₁₆	2 ³ / ₈	3 ³ / ₈	1 ⁵ / ₈	1 ⁵ / ₁₆	3 ¹⁷ / ₃₂	73
16.8	2	50	24	45 ¹³ / ₃₂	65 ⁷ / ₈	52 ¹¹ / ₁₆	2 ¹⁵ / ₃₂	3 ²¹ / ₃₂	1 ³¹ / ₃₂	1 ⁹ / ₁₆	4 ⁷ / ₃₂	110
27.2	2 ¹ / ₂	64	24	49 ¹³ / ₃₂	72 ³ / ₃₂	58 ⁹ / ₃₂	2 ¹⁵ / ₁₆	4 ¹ / ₂	2 ¹ / ₄	1 ⁵ / ₈	5 ⁵ / ₈	194
34	2 ³ / ₄	69	24	53 ¹ / ₁₆	74 ³ / ₃₂	63 ⁵ / ₃₂	3 ¹⁷ / ₃₂	4 ¹¹ / ₃₂	2 ³ / ₄	1 ⁵ / ₈	6 ⁷ / ₃₂	240

CAD INFO



G-6333



Green Pin Polar® JJ Turnbuckle BN

Grade 80 turnbuckle with jaw-jaw end-fitting and safety bolt for use under low temperatures, generally to ASTM F1145-92

- **Material:** drop forged alloy steel, grade 80, quenched and tempered
- **Safety factor:** MBL equals 5 x WLL
- **Standard:** ASME B30.26 generally to ASTM F1145-92
formerly U.S. Federal Specification FF-T-791b
- **Finish:** hot dipped galvanized
- **Temperature range:** -40 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 MTC^a CE IIA
- **Article code:** scan QR code to see article codes
- **Note:** jaw ends up to and including $\frac{5}{8}$ " are fitted with bolts and nuts, sizes $\frac{3}{4}$ " and up are equipped with bolts, nuts and cotter pins



working load limit	diameter thread		take up	length closed position	length open position	length closed position	opening jaw	length inside	diameter pin	thickness jaw eye	diameter jaw eye	weight each
metric t	a inch	a mm	inch	b inch	c inch	d inch	e inch	f inch	g inch	h inch	i inch	lbs
1	$\frac{1}{2}$	12	12	$17 \frac{29}{32}$	$29 \frac{1}{8}$	$19 \frac{7}{16}$	$\frac{5}{8}$	$1 \frac{1}{32}$	$\frac{13}{32}$	$\frac{7}{16}$	$\frac{31}{32}$	3.04
1.59	$\frac{5}{8}$	16	12	$19 \frac{19}{32}$	$30 \frac{15}{32}$	$21 \frac{15}{16}$	$\frac{23}{32}$	$1 \frac{1}{4}$	$\frac{1}{2}$	$\frac{9}{16}$	$1 \frac{5}{16}$	5.11
2.36	$\frac{3}{4}$	19	18	$26 \frac{3}{8}$	$43 \frac{5}{32}$	$29 \frac{1}{8}$	$\frac{15}{16}$	$1 \frac{1}{2}$	$\frac{5}{8}$	$\frac{5}{8}$	$1 \frac{5}{8}$	10.08
3.27	$\frac{7}{8}$	22	18	$28 \frac{1}{16}$	$44 \frac{9}{16}$	$31 \frac{3}{32}$	$1 \frac{1}{16}$	$1 \frac{21}{32}$	$\frac{3}{4}$	$\frac{3}{4}$	$1 \frac{7}{8}$	14.33
4.54	1	25	18	$29 \frac{17}{32}$	$45 \frac{31}{32}$	$32 \frac{7}{8}$	$1 \frac{7}{32}$	$1 \frac{31}{32}$	$\frac{7}{8}$	$\frac{25}{32}$	$2 \frac{1}{8}$	18.52
6.9	$1 \frac{1}{4}$	32	18	$31 \frac{11}{16}$	$48 \frac{7}{16}$	$35 \frac{13}{16}$	$1 \frac{23}{32}$	$2 \frac{25}{32}$	$1 \frac{3}{32}$	$1 \frac{1}{32}$	$2 \frac{11}{16}$	30
9.71	$1 \frac{1}{2}$	38	18	$32 \frac{15}{32}$	$48 \frac{31}{32}$	$37 \frac{5}{8}$	$2 \frac{1}{16}$	$2 \frac{25}{32}$	$1 \frac{3}{8}$	$1 \frac{3}{32}$	$3 \frac{5}{32}$	46.5
12.7	$1 \frac{3}{4}$	45	18	$36 \frac{15}{16}$	$51 \frac{13}{16}$	43	$2 \frac{3}{8}$	$3 \frac{3}{8}$	$1 \frac{5}{8}$	$1 \frac{5}{16}$	$3 \frac{17}{32}$	66

CAD INFO



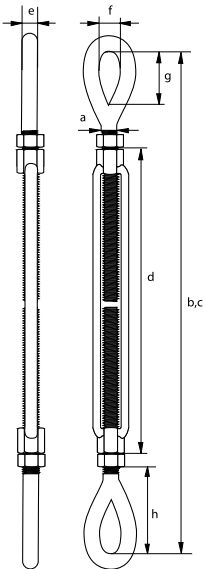
Green Pin® EE Turnbuckle

Turnbuckle with eye-eye end-fitting,
generally to ASTM F1145-92

Scan for
additional
product
details



G-6311



- **Material:** drop forged high tensile steel SAE 1035 or 1045
- **Safety factor:** MBL equals 5 x WLL
- **Standard:** generally to ASTM F1145-92
formerly U.S. Federal Specification FF-T-791b
- **Finish:** hot dipped galvanized
- **Temperature range:** -4 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 MTC³ CE IIA
- **Article code:** scan QR code to see article codes

working load limit	diameter thread		take up	length closed position	length open position	length	diameter	width inside	length inside	length closed position	weight each
metric t	a inch	a mm	inch	b inch	c inch	d inch	e inch	f inch	g inch	h inch	lbs
0.54	3/8	10	6	11 1/2	16 27/32	7 7/32	13/32	1/2	1 5/32	1 15/16	1.06
1	1/2	12	6	12 25/32	17 29/32	7 19/32	15/32	23/32	1 13/32	2 9/32	1.79
1	1/2	12	9	15 3/4	23 15/16	10 5/8	15/32	23/32	1 13/32	2 1/4	2.36
1	1/2	12	12	18 3/4	29 29/32	13 5/8	15/32	23/32	1 13/32	2 1/4	2.84
1.59	5/8	16	6	14 31/32	19 13/16	8	9/16	13/16	1 25/32	3 1/8	2.93
1.59	5/8	16	9	17 29/32	25 13/16	11 1/32	9/16	13/16	1 25/32	3 1/16	3.55
1.59	5/8	16	12	20 29/32	31 13/16	14 1/32	9/16	13/16	1 25/32	3 1/16	4.32
2.36	3/4	19	6	16 1/4	20 15/16	8 7/16	21/32	1 1/32	2 1/8	3 1/2	4.48
2.36	3/4	19	9	19 9/32	26 31/32	11 15/32	21/32	1 1/32	2 1/8	3 1/2	5.45
2.36	3/4	19	12	22 7/32	32 15/16	14 7/16	21/32	1 1/32	2 1/8	3 15/32	6.39
2.36	3/4	19	18	28 9/32	45	20 7/16	21/32	1 1/32	2 1/8	3 1/2	8.69
3.27	7/8	22	12	23 25/32	34 1/4	14 27/32	25/32	1 1/4	2 13/32	3 31/32	9.5
3.27	7/8	22	18	29 3/4	46 7/32	20 13/16	25/32	1 1/4	2 13/32	3 31/32	12.15
4.54	1	25	6	19 19/32	23 25/32	9 7/32	15/16	1 15/32	3	4 21/32	9.33
4.54	1	25	12	25 9/16	35 25/32	15 1/4	15/16	1 15/32	3	4 19/32	12.68
4.54	1	25	18	31 17/32	47 27/32	21 7/32	15/16	1 15/32	3	4 19/32	16.03
4.5	1	25	24	37 15/32	59 3/4	27 1/4	15/16	1 15/32	3	4 9/16	17.2
6.9	1 1/4	32	12	28 1/32	38 25/32	15 5/32	1 5/32	1 27/32	3 19/32	5 23/32	20.5
6.9	1 1/4	32	18	33 15/16	50 21/32	21 5/32	1 5/32	1 27/32	3 19/32	5 21/32	24.5
6.9	1 1/4	32	24	39 31/32	62 11/16	27 5/32	1 5/32	1 27/32	3 19/32	5 21/32	26.7
9.71	1 1/2	38	12	29 3/4	40 9/32	15 25/32	1 1/4	2 5/32	4 3/16	6 5/32	27.8
9.71	1 1/2	38	18	36 1/16	52 9/16	21 25/32	1 1/4	2 5/32	4 3/16	6 5/16	34.8
9.71	1 1/2	38	24	41 15/16	64 13/32	27 25/32	1 1/4	2 5/32	4 3/16	6 7/32	38.1
12.7	1 3/4	45	18	40 5/32	54 31/32	22 23/32	1 1/2	2 13/32	4 23/32	7 3/4	53
12.7	1 3/4	45	24	46 3/32	67 1/16	28 3/4	1 1/2	2 13/32	4 23/32	7 23/32	59
16.8	2	50	24	49 3/4	70 1/4	29 7/16	1 13/16	2 23/32	5 25/32	9 1/16	84
27.2	2 1/2	64	24	56 5/16	76 5/32	31 9/16	2	3 5/32	6 1/2	10 25/32	143
34	2 3/4	69	24	57 3/32	78 9/32	31 9/16	2 1/4	3 5/16	7	11 3/16	185

CAD INFO



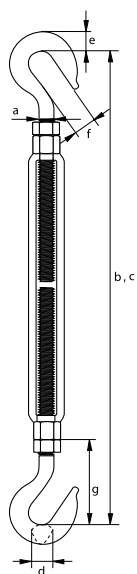
Green Pin® HH Turnbuckle

Turnbuckle with hook-hook end-fitting,
generally to ASTM1145-92

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additional
product
details



G-6312



- **Material:** drop forged high tensile steel SAE 1035 or 1045
- **Safety factor:** MBL equals 5 x WLL
- **Standard:** generally to ASTM F1145-92
formerly U.S. Federal Specification FF-T-791b
hot dipped galvanized
- **Finish:**
- **Temperature range:** -4 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 MTC^a CE IIA
- **Article code:** scan QR code to see article codes

working load limit	diameter thread		take up	length closed position	length open position	thickness hook	thickness hook	opening hook	length closed position	weight each
metric t	a inch	a mm	inch	b inch	c inch	d inch	e inch	f inch	g inch	lbs
0.54	³ / ₈	10	6	10 ¹⁵ / ₁₆	16 ¹¹ / ₃₂	¹ / ₂	⁵ / ₈	¹⁹ / ₃₂	1 ²¹ / ₃₂	1.17
1	¹ / ₂	12	6	12	17 ³ / ₃₂	⁵ / ₈	⁷ / ₈	⁵ / ₈	1 ⁷ / ₈	2.05
1	¹ / ₂	12	9	14 ³¹ / ₃₂	23 ¹ / ₈	⁵ / ₈	⁷ / ₈	⁵ / ₈	1 ²⁷ / ₃₂	2.56
0.68	¹ / ₂	12	12	17 ¹⁵ / ₁₆	29 ³ / ₃₂	¹ / ₂	³ / ₄	⁵ / ₈	1 ²⁷ / ₃₂	2.95
1.59	⁵ / ₈	16	6	14 ¹ / ₃₂	18 ²⁷ / ₃₂	⁵ / ₈	²⁹ / ₃₂	¹³ / ₁₆	2 ⁵ / ₈	3.09
1.59	⁵ / ₈	16	9	16 ³¹ / ₃₂	24 ⁷ / ₈	²⁵ / ₃₂	¹⁵ / ₁₆	¹³ / ₁₆	2 ¹⁹ / ₃₂	4.32
1.59	⁵ / ₈	16	12	19 ³¹ / ₃₂	30 ⁷ / ₈	⁵ / ₈	²⁹ / ₃₂	¹³ / ₁₆	2 ¹⁹ / ₃₂	3.95
2.36	³ / ₄	19	6	15 ¹⁵ / ₃₂	20 ¹ / ₈	⁷ / ₈	1 ¹ / ₁₆	¹⁵ / ₁₆	3 ¹ / ₈	4.5
1.36	³ / ₄	19	9	18 ⁷ / ₁₆	26 ⁵ / ₃₂	²⁵ / ₃₂	1 ¹ / ₁₆	¹⁵ / ₁₆	3 ¹ / ₁₆	5.49
2.36	³ / ₄	19	12	21 ¹³ / ₃₂	32 ¹ / ₈	⁷ / ₈	1 ¹ / ₁₆	¹⁵ / ₁₆	3 ¹ / ₁₆	7.21
4.54	1	25	6	18 ²⁷ / ₃₂	23 ¹ / ₁₆	1 ¹ / ₃₂	1 ³ / ₈	1 ⁷ / ₃₂	4 ⁹ / ₃₂	9.33
4.54	1	25	12	24 ¹⁹ / ₃₂	34 ⁷ / ₈	1 ¹ / ₃₂	1 ³ / ₈	1 ⁷ / ₃₂	4 ³ / ₁₆	14.64
2.27	1	25	18	30 ⁵ / ₈	46 ⁷ / ₈	1 ¹ / ₃₂	1 ³ / ₈	1 ⁷ / ₃₂	4 ³ / ₁₆	14.33
2.27	1	25	24	36 ¹⁷ / ₃₂	58 ²⁷ / ₃₂	1 ¹ / ₃₂	1 ³ / ₈	1 ⁷ / ₃₂	4 ¹ / ₈	17.92

CAD INFO

Green Pin® EJ Turnbuckle

Turnbuckle with eye-jaw end-fitting and cotter pin or safety bolt (depending on size), generally to ASTM1145-92

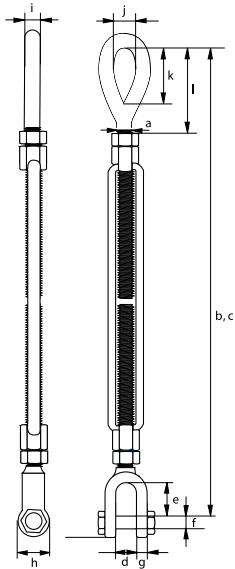
Scan for additional product details



- **Material:** drop forged high tensile steel SAE 1035 or 1045
- **Safety factor:** MBL equals 5 x WLL
- **Standard:** generally to ASTM F1145-92
formerly U.S. Federal Specification FF-T-791b
- **Finish:** hot dipped galvanized
- **Temperature range:** -4 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 MTC³ CE IIA
- **Article code:** scan QR code to see article codes



G-6315



working load limit	diameter thread		take up	length closed position	length open position	opening jaw	length inside jaw	diameter pin jaw	thickness eye jaw	diameter eye jaw	diameter eye	width inside eye	length inside eye	length closed position	weight each
metric t	a inch	a mm	inch	b inch	c inch	d inch	e inch	f inch	g inch	h inch	i inch	j inch	k inch	l inch	lbs
0.54	³ / ₈	10	6	11 ⁵ / ₃₂	16 ¹⁵ / ₃₂	¹⁵ / ₃₂	¹³ / ₁₆	⁵ / ₁₆	¹¹ / ₃₂	¹³ / ₁₆	¹³ / ₃₂	¹ / ₂	1 ⁵ / ₃₂	1 ¹⁵ / ₁₆	1.15
1	¹ / ₂	12	6	12 ¹³ / ₃₂	17 ⁹ / ₁₆	⁵ / ₈	1 ¹ / ₃₂	¹³ / ₃₂	⁷ / ₁₆	³¹ / ₃₂	¹⁵ / ₃₂	²³ / ₃₂	1 ¹³ / ₃₂	2 ⁹ / ₃₂	1.94
1	¹ / ₂	12	9	15 ¹¹ / ₃₂	23 ¹⁷ / ₃₂	⁵ / ₈	1 ¹ / ₃₂	¹³ / ₃₂	⁷ / ₁₆	³¹ / ₃₂	¹⁵ / ₃₂	²³ / ₃₂	1 ¹³ / ₃₂	2 ¹ / ₄	2.49
1	¹ / ₂	12	12	18 ¹¹ / ₃₂	29 ⁹ / ₁₆	⁵ / ₈	1 ¹ / ₃₂	¹³ / ₃₂	⁷ / ₁₆	³¹ / ₃₂	¹⁵ / ₃₂	²³ / ₃₂	1 ¹³ / ₃₂	2 ¹ / ₄	3.02
1.59	⁵ / ₈	16	6	14 ⁹ / ₃₂	19 ¹ / ₈	²³ / ₃₂	1 ¹ / ₄	¹ / ₂	⁹ / ₁₆	1 ⁵ / ₁₆	⁹ / ₁₆	¹³ / ₁₆	1 ²⁵ / ₃₂	3 ¹ / ₈	3.42
1.59	⁵ / ₈	16	9	17 ¹ / ₄	25 ⁵ / ₃₂	²³ / ₃₂	1 ¹ / ₄	¹ / ₂	⁹ / ₁₆	1 ⁵ / ₁₆	⁹ / ₁₆	¹³ / ₁₆	1 ²⁵ / ₃₂	3 ¹ / ₁₆	3.55
1.59	⁵ / ₈	16	12	20 ¹ / ₄	31 ³ / ₃₂	²³ / ₃₂	1 ¹ / ₄	¹ / ₂	⁹ / ₁₆	1 ⁵ / ₁₆	⁹ / ₁₆	¹³ / ₁₆	1 ²⁵ / ₃₂	3 ¹ / ₁₆	4.78
2.36	³ / ₄	19	6	15 ¹³ / ₃₂	20 ³ / ₃₂	¹⁵ / ₁₆	1 ¹ / ₂	⁵ / ₈	⁵ / ₈	1 ⁵ / ₈	²¹ / ₃₂	1 ¹ / ₃₂	2 ¹ / ₈	3 ¹ / ₂	5.03
2.36	³ / ₄	19	9	18 ³ / ₈	26 ³ / ₃₂	¹⁵ / ₁₆	1 ¹ / ₂	⁵ / ₈	⁵ / ₈	1 ⁵ / ₈	²¹ / ₃₂	1 ¹ / ₃₂	2 ¹ / ₈	3 ¹ / ₂	6.22
2.36	³ / ₄	19	12	21 ¹¹ / ₃₂	32 ³ / ₃₂	¹⁵ / ₁₆	1 ¹ / ₂	⁵ / ₈	⁵ / ₈	1 ⁵ / ₈	²¹ / ₃₂	1 ¹ / ₃₂	2 ¹ / ₈	3 ¹⁵ / ₃₂	6.97
2.36	³ / ₄	19	18	27 ⁵ / ₁₆	44 ³ / ₃₂	¹⁵ / ₁₆	1 ¹ / ₂	⁵ / ₈	⁵ / ₈	1 ⁵ / ₈	²¹ / ₃₂	1 ¹ / ₃₂	2 ¹ / ₈	3 ¹ / ₂	9.04
3.27	⁷ / ₈	22	12	22 ¹⁵ / ₁₆	33 ³ / ₈	1 ¹ / ₁₆	1 ²¹ / ₃₂	³ / ₄	³ / ₄	1 ⁷ / ₈	²⁵ / ₃₂	1 ¹ / ₄	2 ¹³ / ₃₂	3 ³¹ / ₃₂	9.04
3.27	⁷ / ₈	22	18	28 ¹⁵ / ₁₆	45 ¹³ / ₃₂	1 ¹ / ₁₆	1 ²¹ / ₃₂	³ / ₄	³ / ₄	1 ⁷ / ₈	²⁵ / ₃₂	1 ¹ / ₄	2 ¹³ / ₃₂	3 ³¹ / ₃₂	12.87
4.54	1	25	6	18 ⁵ / ₈	22 ²⁵ / ₃₂	1 ⁷ / ₃₂	1 ³¹ / ₃₂	⁷ / ₈	²⁵ / ₃₂	2 ¹ / ₈	¹⁵ / ₁₆	1 ¹⁵ / ₃₂	3	4 ²¹ / ₃₂	10.14
4.54	1	25	12	24 ⁵ / ₈	34 ¹³ / ₁₆	1 ⁷ / ₃₂	1 ³¹ / ₃₂	⁷ / ₈	²⁵ / ₃₂	2 ¹ / ₈	¹⁵ / ₁₆	1 ¹⁵ / ₃₂	3	4 ¹⁹ / ₃₂	13.6
4.54	1	25	18	30 ⁹ / ₁₆	46 ²⁷ / ₃₂	1 ⁷ / ₃₂	1 ³¹ / ₃₂	⁷ / ₈	²⁵ / ₃₂	2 ¹ / ₈	¹⁵ / ₁₆	1 ¹⁵ / ₃₂	3	4 ¹⁹ / ₃₂	15.65
4.5	1	25	24	36 ¹⁷ / ₃₂	58 ¹³ / ₁₆	1 ⁷ / ₃₂	1 ³¹ / ₃₂	⁷ / ₈	²⁵ / ₃₂	2 ¹ / ₈	¹⁵ / ₁₆	1 ¹⁵ / ₃₂	3	4 ⁹ / ₁₆	18.41
6.9	1 ¹ / ₄	32	12	26 ²¹ / ₃₂	37 ¹³ / ₃₂	1 ²³ / ₃₂	2 ²⁵ / ₃₂	1 ⁵ / ₃₂	1 ¹ / ₃₂	2 ¹¹ / ₁₆	1 ⁵ / ₃₂	1 ²⁷ / ₃₂	3 ¹⁹ / ₃₂	5 ²³ / ₃₂	22.9
6.9	1 ¹ / ₄	32	18	32 ²⁵ / ₃₂	49 ¹⁷ / ₃₂	1 ²³ / ₃₂	2 ²⁵ / ₃₂	1 ⁵ / ₃₂	1 ¹ / ₃₂	2 ¹¹ / ₁₆	1 ⁵ / ₃₂	1 ²⁷ / ₃₂	3 ¹⁹ / ₃₂	5 ²¹ / ₃₂	25.4
6.9	1 ¹ / ₄	32	24	38 ¹⁵ / ₁₆	61 ²¹ / ₃₂	1 ²³ / ₃₂	2 ²⁵ / ₃₂	1 ⁵ / ₃₂	1 ¹ / ₃₂	2 ¹¹ / ₁₆	1 ⁵ / ₃₂	1 ²⁷ / ₃₂	3 ¹⁹ / ₃₂	5 ²¹ / ₃₂	29.1
9.71	1 ¹ / ₂	38	12	28 ³ / ₁₆	38 ¹¹ / ₁₆	2 ¹ / ₁₆	2 ²⁵ / ₃₂	1 ³ / ₈	1 ¹ / ₃₂	3 ⁵ / ₃₂	1 ¹ / ₄	2 ⁵ / ₃₂	4 ³ / ₁₆	6 ⁵ / ₃₂	30.6
9.71	1 ¹ / ₂	38	18	34 ⁹ / ₃₂	50 ²⁵ / ₃₂	2 ¹ / ₁₆	2 ²⁵ / ₃₂	1 ³ / ₈	1 ¹ / ₃₂	3 ⁵ / ₃₂	1 ¹ / ₄	2 ⁵ / ₃₂	4 ³ / ₁₆	6 ⁵ / ₁₆	36.8
9.71	1 ¹ / ₂	38	24	40 ⁹ / ₃₂	62 ³ / ₄	2 ¹ / ₁₆	2 ²⁵ / ₃₂	1 ³ / ₈	1 ¹ / ₃₂	3 ⁵ / ₃₂	1 ¹ / ₄	2 ⁵ / ₃₂	4 ³ / ₁₆	6 ⁷ / ₃₂	43.4
12.7	1 ³ / ₄	45	18	38 ¹⁷ / ₃₂	53 ³ / ₈	2 ³ / ₈	3 ³ / ₈	1 ⁵ / ₈	1 ⁵ / ₁₆	3 ¹⁷ / ₃₂	1 ¹ / ₂	2 ¹³ / ₃₂	4 ²³ / ₃₂	7 ³ / ₄	55
12.7	1 ³ / ₄	45	24	44 ¹ / ₂	65 ⁷ / ₁₆	2 ³ / ₈	3 ³ / ₈	1 ⁵ / ₈	1 ⁵ / ₁₆	3 ¹⁷ / ₃₂	1 ¹ / ₂	2 ¹³ / ₃₂	4 ²³ / ₃₂	7 ²³ / ₃₂	64
16.8	2	50	24	47 ⁹ / ₁₆	68 ¹ / ₃₂	2 ¹⁵ / ₃₂	3 ²¹ / ₃₂	2	1 ⁹ / ₁₆	4 ⁷ / ₃₂	1 ¹³ / ₁₆	2 ²³ / ₃₂	5 ²⁵ / ₃₂	9 ¹ / ₁₆	93
27.2	2 ¹ / ₂	64	24	52 ⁷ / ₈	74 ³ / ₄	2 ¹⁵ / ₁₆	4 ¹ / ₂	2 ¹ / ₄	1 ⁵ / ₈	5 ⁵ / ₈	2	3 ⁵ / ₃₂	6 ¹ / ₂	10 ²⁵ / ₃₂	151
34	2 ³ / ₄	69	24	55 ³ / ₃₂	76 ⁷ / ₈	3 ¹⁷ / ₃₂	4 ¹¹ / ₃₂	2 ³ / ₄	1 ⁵ / ₈	6 ⁷ / ₃₂	2 ¹ / ₄	3 ⁵ / ₁₆	7	11 ³ / ₁₆	205

CAD INFO



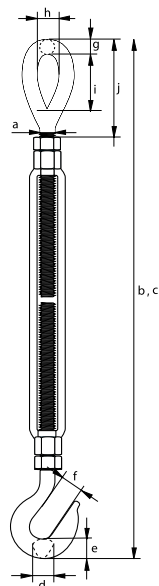
Green Pin® EH Turnbuckle

Turnbuckle with eye-hook end-fitting,
generally to ASTM F1145-92

Scan for
additional
product
details



G-6314



- **Material:** drop forged high tensile steel SAE 1035 or 1045
- **Safety factor:** MBL equals 5 x WLL
- **Standard:** generally to ASTM F1145-92
formerly U.S. Federal Specification FF-T-791b
hot dipped galvanized
- **Finish:** hot dipped galvanized
- **Temperature range:** -4 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 MTC^a CE II A
- **Article code:** scan QR code to see article codes

working load limit	diameter thread		take up	length closed position	length open position	thickness hook	thickness hook	opening hook	diameter eye	width inside eye	length inside eye	length closed position	weight each
metric t	a inch	a mm	inch	b inch	c inch	d inch	e inch	f inch	g inch	h inch	i inch	j inch	lbs
0.54	³ / ₈	10	6	11 ⁷ / ₃₂	16 ⁵ / ₈	¹ / ₂	⁵ / ₈	¹⁹ / ₃₂	¹³ / ₃₂	¹ / ₂	1 ⁵ / ₃₂	1 ¹⁵ / ₁₆	0.88
1	¹ / ₂	12	6	12 ¹³ / ₃₂	17 ¹⁷ / ₃₂	⁵ / ₈	⁷ / ₈	⁵ / ₈	¹⁵ / ₃₂	²³ / ₃₂	1 ¹³ / ₃₂	2 ⁹ / ₃₂	1.61
1	¹ / ₂	12	9	15 ¹¹ / ₃₂	23 ¹⁷ / ₃₂	⁵ / ₈	⁷ / ₈	⁵ / ₈	¹⁵ / ₃₂	²³ / ₃₂	1 ¹³ / ₃₂	2 ¹ / ₄	1.96
0.68	¹ / ₂	12	12	18 ¹¹ / ₃₂	29 ¹⁷ / ₃₂	¹ / ₂	³ / ₄	⁵ / ₈	¹⁵ / ₃₂	²³ / ₃₂	1 ¹³ / ₃₂	2 ¹ / ₄	2.31
1.59	⁵ / ₈	16	6	14 ¹ / ₂	19 ¹¹ / ₃₂	⁵ / ₈	²⁹ / ₃₂	¹³ / ₁₆	⁹ / ₁₆	¹³ / ₁₆	1 ²⁵ / ₃₂	3 ¹ / ₈	2.45
1.59	⁵ / ₈	16	9	17 ⁷ / ₁₆	25 ¹¹ / ₃₂	²⁵ / ₃₂	¹⁵ / ₁₆	¹³ / ₁₆	⁹ / ₁₆	¹³ / ₁₆	1 ²⁵ / ₃₂	3 ¹ / ₁₆	3.06
1.59	⁵ / ₈	16	12	20 ⁷ / ₁₆	31 ¹¹ / ₃₂	⁵ / ₈	²⁹ / ₃₂	¹³ / ₁₆	⁹ / ₁₆	¹³ / ₁₆	1 ²⁵ / ₃₂	3 ¹ / ₁₆	3.95
2.36	³ / ₄	19	6	15 ⁷ / ₈	20 ¹ / ₂	⁷ / ₈	1 ¹ / ₁₆	¹⁵ / ₁₆	²¹ / ₃₂	1 ¹ / ₃₂	2 ¹ / ₈	3 ¹ / ₂	4.03
1.36	³ / ₄	19	9	18 ²⁷ / ₃₂	26 ⁹ / ₁₆	²⁵ / ₃₂	1 ¹ / ₁₆	¹⁵ / ₁₆	²¹ / ₃₂	1 ¹ / ₃₂	2 ¹ / ₈	3 ¹ / ₂	5.07
2.36	³ / ₄	19	12	21 ¹³ / ₁₆	32 ⁹ / ₁₆	⁷ / ₈	1 ¹ / ₁₆	¹⁵ / ₁₆	²¹ / ₃₂	1 ¹ / ₃₂	2 ¹ / ₈	3 ¹⁵ / ₃₂	5.07
4.54	1	25	6	19 ⁷ / ₃₂	23 ⁷ / ₁₆	1 ¹ / ₃₂	1 ³ / ₈	1 ⁷ / ₃₂	¹⁵ / ₁₆	1 ¹³ / ₃₂	2 ¹⁵ / ₁₆	4 ²¹ / ₃₂	8.6
4.54	1	25	12	25 ¹ / ₃₂	35 ⁵ / ₁₆	1 ¹ / ₃₂	1 ³ / ₈	1 ⁷ / ₃₂	¹⁵ / ₁₆	1 ¹³ / ₃₂	2 ¹⁵ / ₁₆	4 ¹⁹ / ₃₂	12.08
2.27	1	25	18	31 ¹ / ₁₆	47 ⁵ / ₁₆	1 ¹ / ₃₂	1 ³ / ₈	1 ⁷ / ₃₂	¹⁵ / ₁₆	1 ¹³ / ₃₂	2 ¹⁵ / ₁₆	4 ¹⁹ / ₃₂	13.23
2.27	1	25	24	36 ³¹ / ₃₂	59 ⁹ / ₃₂	1 ¹ / ₃₂	1 ³ / ₈	1 ⁷ / ₃₂	¹⁵ / ₁₆	1 ¹³ / ₃₂	2 ¹⁵ / ₁₆	4 ⁹ / ₁₆	16.58

CAD INFO



CM[®] **CHAIN & RIGGING ATTACHMENTS**

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C O L U M B U S M c K I N N O N C O R P O R A T I O N



TURNBUCKLES

WORKING LOAD LIMIT: 400 TO 21,400 LBS.

BENEFITS & FEATURES

- Turnbuckles can be used to apply tension to wire rope or cable.
- Composed of a forged body and two end fittings. End fittings can be eyes, jaws, or hooks.
- Must only be applied to loads within their working load limit as specified for the particular end fitting being applied.
- Apply turnbuckles in a straight in-line manner only. Do not allow anything to contact the turnbuckle body or end attachment threaded shanks.
- Apply load to the center of end attachment eyes and bowl of hooks. Do not tip load hooks or side load eyes.
- Design factor 5:1
- ASTMF-1145



Eye & Eye



Jaw & Eye



Jaw & Jaw



Hook & Eye



Hook & Hook

CARE & INSPECTION

Inspect turnbuckles before use for bent components and worn threads. Do not use if body or end fitting is bent more than 10° from the axial center line. Do not use if threads are visibly worn or feel loose. If in doubt, consult a rigging handbook or discuss with a qualified person.

WARNING

Improper use or care of turnbuckles can result in bodily injury or property damage. To avoid injury:

- Inspect turnbuckles for distortion and wear.
- Do not use if anything is in contact with the turnbuckle body or an end fitting.
- Only apply load to center of eyes and bowl of hooks.
- Do not exceed working load limit.

TURNBUCKLE LOCK NUTS



Size (in.)	Product Code		Weight (lbs.)
	Right Hang Nut	Left Hang Nut	
1/4	3X845	3X845L	0.005
5/16	—	—	0.008
3/8	3X847	—	0.010
1/2	3X849R	3X849L	0.030
5/8	—	3X850L	0.050
3/4	—	3X851L	0.090
7/8	—	3X852L	0.130
1	—	3X853L	0.200
1-1/4	—	3X855L	0.410
1-1/2	—	3X866L	0.700

• Imported. Hot Dip Galvanized.

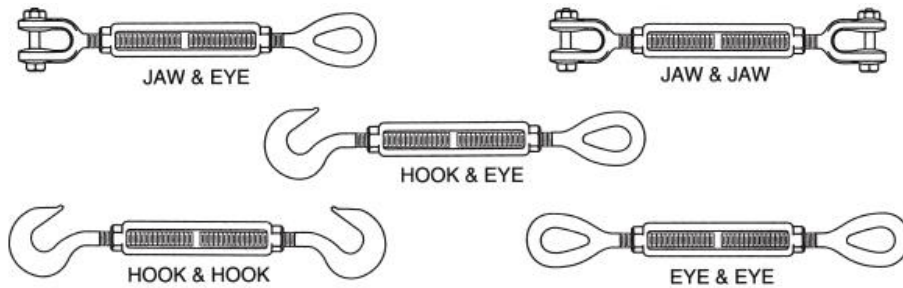
Eye & Eye		Jaw & Eye		Jaw & Jaw		Thread Diameter (in.)	Take Up (in.)	WLL (lbs.)	Weight (lbs.)
Product Code	Weight (lbs.)	Product Code	Weight (lbs.)	Product Code	Weight (lbs.)				
0404EE	0.3	—	—	0404JJ	0.4	1/4	4	500	0.4
0504EE	0.5	—	—	0504JJ	0.6	5/16	4.5	700	0.6
0606EE	0.8	0606JE	0.9	0606JJ	0.9	3/8	6	1,040	0.9
0806EE	1.5	0806JE	1.7	0806JJ	1.8	1/2	6	1,800	1.8
0809EE	1.9	—	2.0	0809JJ	2.1	1/2	9	1,800	2.1
0812EE	2.3	0812JE	2.4	0812JJ	2.4	1/2	12	1,800	2.4
1006EE	2.5	1006JE	3.0	1006JJ	3.0	5/8	6	2,700	3.0
1009EE	3.3	—	—	1009JJ	3.7	5/8	9	2,700	3.7
1012EE	3.8	1012JE	4.0	1012JJ	4.2	5/8	12	2,700	4.2
1206EE	3.9	1206JE	4.3	1206JJ	4.6	3/4	6	4,000	4.6
1209EE	4.8	1209JE	5.1	1209JJ	5.4	3/4	9	4,000	5.4
1212EE	5.4	1212JE	5.7	1212JJ	6.0	3/4	12	4,000	6.0
1218EE	7.0	1218JE	7.3	1218JJ	7.7	3/4	18	4,000	7.7
1412EE	7.4	1412JE	7.9	1412JJ	8.4	7/8	12	5,800	8.4
1418EE	9.6	1418JE	10.2	1418JJ	10.7	7/8	18	5,800	10.7
1606EE	9.0	1606JE	9.4	1606JJ	9.7	1	6	7,600	9.7
1612EE	11.2	1612JE	11.6	1612JJ	11.9	1	12	7,600	11.9
1618EE	13.8	1618JE	14.3	1618JJ	14.8	1	18	7,600	14.8
—	—	1624JE	17.6	1624JJ	18.2	1	24	7,600	18.2
2012EE	20.9	—	—	2012JJ	23.8	1-1/4	12	12,000	23.8
2018EE	25.7	—	—	2018JJ	27.5	1-1/4	18	12,000	27.5
—	—	—	—	2024JJ	33.7	1-1/4	24	12,000	33.7
2412EE	29	—	—	2412JJ	38.1	1-1/2	12	17,000	38.1
—	—	—	—	2418JJ	44.2	1-1/2	18	17,000	44.2
2424EE	40.7	—	—	2424JJ	48.4	1-1/2	24	17,000	48.4

• Imported. Hot Dip Galvanized.

Product Code		Thread Diameter (in.)	Take Up (in.)	WLL (lbs.)	Weight (lbs.)
Hook & Eye	Hook & Hook				
0404HE	0404HH	1/4	4	300	—
0806HE	0806HH	1/2	6	1,040	1.5
—	0809HH	1/2	9	1,040	1.9
0812HE	0812HH	1/2	12	1,040	2.3
—	1006HH	5/8	6	1,600	2.5
1009HE	—	5/8	9	1,600	3.3
—	1012HH	5/8	12	1,600	3.8
—	—	3/4	6	2,000	3.9
1212HE	1212HH	3/4	12	2,000	5.4
—	—	7/8	12	2,400	7.4
—	—	1	6	2,900	9.0
1612HE	1612HH	1	12	2,900	11.2
1618HE	—	1	18	2,900	13.8
—	—	1	24	2,900	17.1

• Imported. Hot Dip Galvanized.

TURNBUCKLES



Size in Inches	Approximate Weight Each in Pounds			Working Load Limit in Lbs.		Average Overall Length with Ends in Closed Position
Diameter x Take-Up	Eyes and/or Hooks	Jaw & Eye	Jaw & Jaw	Hook & Hook	Eye & Eye	
1/4 x 4	0.30	0.30	0.40	400	500	8.25
5/16 x 4.1/2	0.50	0.53	0.58	700	800	9.56
3/8 x 6	0.75	0.82	0.93	1,000	1,200	11.88
1/2 x 6	1.50	1.62	1.68	1,500	2,200	13.31
1/2 x 9	1.75	1.82	1.85	1,500	2,200	16.31
1/2 x 12	2.18	2.19	2.20	1,500	2,200	19.31
5/8 x 6	2.63	2.69	2.82	2,250	3,500	15.30
5/8 x 9	3.00	3.01	3.25	2,250	3,500	18.50
5/8 x 12	3.25	3.50	3.75	2,250	3,500	21.50
3/4 x 6	3.75	4.25	4.68	3,000	5,200	17.00
3/4 x 9	4.50	5.00	5.38	3,000	5,200	20.00
3/4 x 12	5.75	5.75	6.12	3,000	5,200	23.00
3/4 x 18	7.00	7.25	7.75	3,000	5,200	29.00
7/8 x 12	8.38	8.88	9.38	4,000	7,200	24.63
7/8 x 18	10.25	10.60	11.44	4,000	7,200	30.63
1 x 6	10.75	11.10	11.80	5,000	10,000	20.63
1 x 12	11.25	12.00	12.88	5,000	10,000	26.63
1 x 18	14.00	14.75	16.10	5,000	10,000	32.63
1 x 24	17.00	17.75	18.60	5,000	10,000	38.63
1.1/4 x 12	19.00	21.20	23.60	6,500	15,200	29.88
1.1/4 x 18	24.10	26.00	26.60	6,500	15,200	35.88
1.1/4 x 24	25.00	28.70	31.20	6,500	15,200	41.88
1.1/2 x 12	27.00	31.10	35.50	7,500	21,400	32.38
1.1/2 x 18	31.20	36.40	40.70	7,500	21,400	38.38
1.1/2 x 24	38.20	44.20	47.60	7,500	21,400	44.38
1.3/4 x 18	53.00	57.50	62.00	**	28,000	41.38
1.3/4 x 24	58.00	60.00	64.00	**	28,000	47.75
2 x 24	90.00	102.00	115.00	**	37,000	51.75
2.1/2 x 24	163.00	180.00	200.00	**	60,000*	58.50
2.3/4 x 24	180.00	214.00	248.00	**	75,000*	61.50

* Supplied with locknuts.

** Hooks not supplied on sizes larger than 1.1/2" diameter.

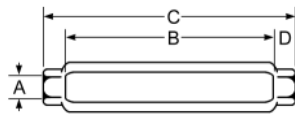
Meets or exceeds performance requirements of A.S.T.M. F 1145. Hot galvanized, drop forged.

Bodies normalized, end fittings quenched and tempered.

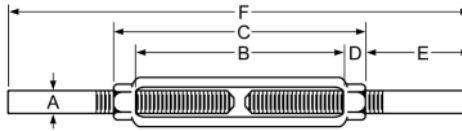
Jaw ends up to 5/8": Fitted with bolt and nut. Jaw ends 3/4" & up: Fitted with pin and cotter.



TURNBUCKLE BODIES / STUB END TB'S



BODY ONLY



STUB END

Size in Inches Dia. (A) x Take-Up (B)	Working Load Limit in Pounds	Approximate Weight Each Body Only in Pounds	Approximate Weight Each Stub End in Pounds	Dimensions in Inches			
				C	D	E	F*
3/8 x 6	1,200	0.3	0.7	7.13	0.56	4.44	16.00
1/2 x 6	2,200	0.6	1.4	7.50	0.75	4.25	16.00
1/2 x 9	2,200	0.8	1.7	10.50	0.75	4.25	19.00
1/2 x 12	2,200	1.0	1.8	13.50	0.75	4.25	22.00
5/8 x 6	3,500	1.0	2.0	7.88	0.94	4.06	16.00
5/8 x 9	3,500	1.2	2.5	10.88	0.94	4.06	19.00
5/8 x 12	3,500	1.4	3.1	13.88	0.94	4.06	22.00
3/4 x 6	5,200	1.2	3.3	8.25	1.13	4.38	17.00
3/4 x 9	5,200	1.6	4.0	11.25	1.13	4.38	20.00
3/4 x 12	5,200	2.1	4.8	14.25	1.13	4.38	23.00
3/4 x 18	5,200	3.2	5.8	20.25	1.13	4.38	29.00
7/8 x 6	7,200	1.8	4.5	8.63	1.31	4.69	18.00
7/8 x 12	7,200	2.7	6.4	14.63	1.31	4.69	24.00
7/8 x 18	7,200	3.8	---	20.63	1.31	---	---
1 x 6	10,000	2.5	6.3	9.00	1.50	5.00	19.00
1 x 12	10,000	4.2	8.9	15.00	1.50	5.00	25.00
1 x 18	10,000	7.0	11.4	21.00	1.50	5.00	31.00
1 x 24	10,000	7.4	---	27.00	1.50	---	---
1.1/8 x 6	12,400	5.0	---	9.13	1.56	---	---
1.1/8 x 12	12,400	5.4	---	15.13	1.56	---	---
1.1/4 x 6	15,200	5.0	10.2	9.13	1.56	5.44	20.00
1.1/4 x 12	15,200	5.4	13.6	15.13	1.56	5.44	26.00
1.1/4 x 18	15,200	8.0	---	21.10	1.56	---	---
1.1/4 x 24	15,200	10.2	---	27.13	1.56	---	---
1.3/8 x 6	18,000	6.0	13.9	9.75	1.88	5.38	20.50
1.1/2 x 6	21,400	5.7	15.1	9.75	1.88	5.38	20.50
1.1/2 x 12	21,400	8.5	20.5	15.75	1.88	5.38	26.50
1.1/2 x 18	21,400	11.0	---	21.75	1.88	---	---
1.1/2 x 24	21,400	13.7	---	27.75	1.88	---	---
1.5/8 x 6	25,000	8.2	---	10.38	2.19	---	---
1.3/4 x 6	28,000	10.3	22.5	10.38	2.19	5.80	22.00
2 x 6	37,000	14.2	34.0	11.00	2.50	6.00	23.00
2 x 24	37,000	30.0	---	29.00	2.50	---	---
2.1/2 x 6	60,000	33.0	68.0	13.50	3.75	6.25	26.00

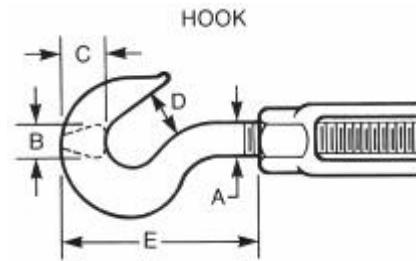
* Closed position

Meets the performance requirements of ASTM F 1145.

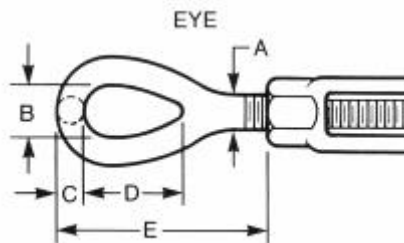
Self-colored finish. Some bodies available hot-galvanized. Drop forged.



TURNBUCKLE END FITTING DIMENSIONS

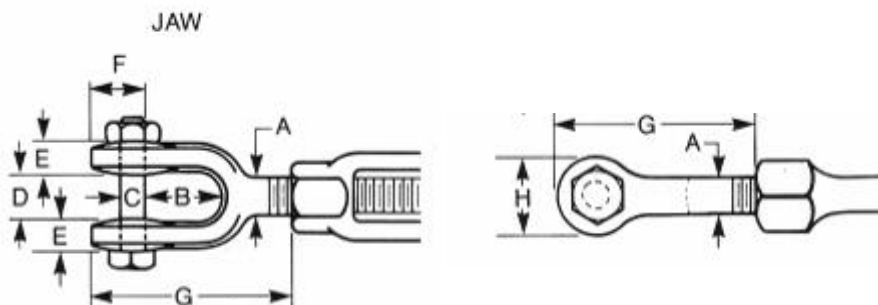


Dimensions in Inches				
A	B	C	D	E
1/4	0.25	0.41	0.45	1.66
5/16	0.31	0.44	0.50	1.91
3/8	0.38	0.53	0.56	2.28
1/2	0.50	0.69	0.66	2.84
5/8	0.63	0.84	0.84	3.53
3/4	0.75	1.00	0.97	4.31
7/8	0.88	1.19	1.13	5.00
1	1.00	1.38	1.25	5.69
1.1/4	1.06	1.50	1.50	6.91
1.1/2	1.31	1.75	1.88	8.69



Dimensions in Inches				
A	B	C	D	E
1/4	0.34	0.22	0.78	1.78
5/16	0.44	0.28	0.94	2.19
3/8	0.53	0.34	1.13	2.56
1/2	0.72	0.44	1.44	3.22
5/8	0.88	0.50	1.75	3.88
3/4	1.00	0.63	2.13	4.69
7/8	1.25	0.75	2.38	5.25
1	1.44	0.88	3.00	6.38
1.1/4	1.81	1.13	3.56	7.75
1.1/2	2.13	1.25	4.13	8.63
1.3/4	2.38	1.50	4.69	10.00
2	2.69	1.75	5.75	12.13
2.1/2	3.13	2.00	6.50	13.56
2.3/4	3.25	2.25	7.00	15.00





Dimensions in Inches							
A	B	C	D	E	F	G	H
1/4	0.63	0.25	0.41	0.28	0.50	1.63	0.63
5/16	0.88	0.25	0.47	0.28	0.50	2.00	0.69
3/8	0.88	0.31	0.50	0.31	0.59	2.19	0.81
1/2	1.06	0.38	0.63	0.41	0.75	2.75	1.00
5/8	1.31	0.50	0.75	0.50	1.03	3.50	1.31
3/4	1.50	0.63	0.94	0.56	1.28	4.13	1.63
7/8	1.75	0.75	1.13	0.69	1.47	4.84	1.88
1	2.06	0.88	1.19	0.78	1.66	5.53	2.13
1.1/4	2.81	1.13	1.75	1.00	2.09	7.19	2.63
1.1/2	2.81	1.38	2.06	1.06	2.47	7.88	3.13
1.3/4	3.38	1.63	2.38	1.25	2.91	9.38	3.50
2	3.75	2.00	2.50	1.56	3.53	10.88	4.19
2.1/2	4.19	2.25	2.88	1.56	4.63	13.34	5.63
2.3/4	4.44	2.75	3.50	1.63	5.38	15.00	6.13

